Financing Innovation: The Role of Patent Examination

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Abstract

We examine how patent system design shapes innovative firms' access to finance. We exploit Britain's 1905 introduction of prior-art search examination, which replaced registration with limited state certification producing public information about patent claims. Using newly compiled data on listed corporations, we find that patenting firms increased borrowing after the reform. The increase was concentrated in bonds, with no comparable rise in bank borrowing, and was accompanied by firm growth. Our results show how low-intensity, low-cost state examination can strengthen patents' informational value, expand innovative firms' access to capital markets, and support the commercialisation and further development of their patented inventions.

Keywords: patent examination, state certification, information asymmetries, financing innovation, bond markets, economic history.

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1 Introduction

Can low-cost state certification improve firms' access to finance? We study Britain's 1905 introduction of prior-art search examination. The reform created a standardised public record about patent claims without changing the legal criteria for patentability or introducing full substantive examination. Following the reform, patenting firms grew and shifted their financing mix towards arm's-length bonds and away from common and preference equity; bank borrowing and trade credit did not respond comparably.

Financing frictions are especially acute for innovative firms. Inventions are intangible, uncertain, and difficult for outsiders to evaluate, leaving firms with few conventional assets against which to raise finance (Hall, 2002; Carpenter and Petersen, 2002; Brown et al., 2009; Hall and Lerner, 2010; Kerr and Nanda, 2015). Long and unpredictable returns make innovative firms still harder for outsiders to evaluate, generating adverse selection and moral hazard problems that can constrain investment and firm growth (Nanda and Rhodes-Kropf, 2016; Hall and Helmers, 2024). Across historical and contemporary settings, these frictions have pushed innovative firms towards more expensive or specialised sources of finance (Nicholas, 2011b; Bakker, 2013). State certification offers one institutional response. By producing credible public information about assets that are otherwise costly to verify, the state may reduce information asymmetries between firms and outsiders.

The patent system provides a useful setting in which to study this response because examination can give otherwise difficult-to-evaluate claims a form of state certification. Patents' value to investors can arise through three functions. First, exclusionary rights may protect expected cash flows by limiting imitation and allowing innovators to recoup their investment (Arrow, 1962; Mansfield et al., 1981; Edquist and Johnson, 1997; Scotchmer, 2004). Second, patents create legally identifiable intangible assets with potential pledgeability or recovery value (Long, 2002; Mann, 2018; Hochberg et al., 2018; Suh, 2023). Third, patent records provide observable evidence of innovative activity, reducing information asymmetries (Hall, 2009; Harhoff, 2009; Hsu and Ziedonis, 2013; Haeussler et al., 2014; Farre-Mensa et al., 2020). These functions are connected: screening and certification can change investors' assessments of the invention, its protected cash flows, and the patent's pledgeable value.

Britain's introduction of patent examination provides a setting in which to examine this certification channel. Taking effect on 1 January 1905, the Patents Act 1902 added a prior-art search step to the patent application process. Trained examiners searched earlier British specifications, and applicants could revise their claims or withdraw before paying a nominal

£1 sealing fee. Because the fee was payable only after the report was issued, it could not meaningfully deter initial applications, although the prospect of scrutiny could discourage applicants expecting adverse findings. The reform introduced a limited, standardised public assessment of patent claims rather than full substantive examination.[1]

This intermediate regime remains relevant because contemporary patent systems also range from registration to prior-art search and full substantive examination (Table 1). The intensity of scrutiny can also vary within these categories with the time, expertise, and resources devoted to each application.[2] Using a calibrated model, Freilich et al. (2024) show how stronger examination can screen out low-quality applications and discourage their submission. As they note, 'understanding the magnitude of a policy change can be critical to determine whether implementing the policy is worth the cost' (Freilich et al., 2024, p. 391). The 1905 reform lets us quantify one potential benefit relevant to that calculation: the extent to which public information from limited, low-cost screening expanded firms' access to external finance.

[Table 1 about here]

We develop a parsimonious framework linking the move from registration towards state certification to external finance. Firms know more than investors about their inventions and patent claims, and under registration investors struggle to distinguish strong from weak invention-patent pairs despite firms' costly and noisy signals. Adapting Schankerman and Schuett (2022), examination can improve the average quality of the patent pool through anticipatory non-entry and post-search amendment or withdrawal. The resulting patents carry standardised certification that their claims have undergone prior-art scrutiny. Following Lemley and Shapiro (2005), our framework treats patents as probabilistic property rights; certification updates investors' assessments of the invention and the probability of effective patent protection, raising expected pledgeable value and the firm's bond capacity.

In our framework, the financing stage follows examination and certification.[3] Public certification should be particularly valuable to dispersed bond investors, who cannot reproduce relationship banks' private information. By partly offsetting this disadvantage, certification may enable firms to place more bonds at prevailing coupons and contractual terms. The framework predicts greater bond capacity and growth where that capacity had constrained

[1] Examiners did not acquire a general power to reject applications on substantive patentability grounds until the Patents Act 1949.

[2] Although the United States formally operates full substantive examination, examiners have historically been allocated just 18 hours per application (Frakes and Wasserman, 2019).

[3] This contrasts with Schuett (2013a), who studies external finance before R&D begins: anticipated examination relaxes an *ex ante* moral-hazard constraint and increases the funding available for research.

profitable investment. Whether firms choose bonds over equity depends on their relative costs, as certification may also reduce the information discount on new equity.

We test these predictions in a difference-in-differences framework. The reform's timing was fixed nationally, and the new process applied to applications reaching the complete-specification stage from 1 January 1905. We interpret the treatment as an improvement in the public information available to external investors about firms' innovation. We construct a new dataset comprising the population of patenting firms officially listed on the London Stock Exchange during 1899–1913. Because no company-level financial database exists for this period, we hand-collected and digitised balance sheets, profit and loss accounts, and corporate governance information from historical archives, following Braggion and Moore (2011). We compare changes in patenting firms' financing and size with a control group of non-patenting firms, using firm and year fixed effects, matching, reweighting, pre-trend tests, and placebo exercises.

Firms patenting after 1905 had debt-to-assets ratios that were 0.024 to 0.027 higher, equivalent to 9.6 to 10.8 per cent of the pre-reform mean. The effect was driven by bond debt, which increased by 0.029 to 0.031 relative to assets, or 18.0 to 19.3 per cent of its pre-reform mean. Bank lending and trade credit did not increase comparably, and bond coupon rates did not decline, consistent with expanded market capacity rather than repricing. Firms with examined patents also relied less on equity, consistent with pecking-order considerations (Myers and Majluf, 1984), and grew by 6.9 to 12.0 per cent. The additional bond finance therefore appears to have supported growth rather than simply rebalanced capital structures. These findings are robust across alternative specifications.

Our contributions are threefold. First, we advance research on innovation finance. Existing work links financial institutions, intangible capital, and corporate debt to innovation and growth (Brown et al., 2013; Hsu et al., 2014; Horsch et al., 2021; Geelen et al., 2022), and studies how innovative assets and patents shape venture selection and startup outcomes (Baum and Silverman, 2004; Hegde et al., 2022). Patents may facilitate finance by disclosing information, protecting inventions, supporting collateralisation, or reflecting innovation quality (Haeussler et al., 2014; Hottenrott et al., 2017; Hochberg et al., 2018; Hall, 2019; Farre-Mensa et al., 2020). We show that their signalling value depends on the institutional process through which claims are assessed. Public screening provides state certification that makes those claims more credible, strengthening patents as signals and potentially pledgeable assets. The bond-market response connects to evidence that public patent information affects external credit (Saidi and Žaldokas, 2021).[4] Our analysis also complements work on when patent information becomes

[4]Gill and Heller (2024) show that a 2004 EU reform strengthening patent rights increased long-term debt

public (Hegde et al., 2023), while we examine its credibility. More broadly, the findings show how low-intensity screening can make technical intangible assets more legible to outsiders, informing debates over financing innovation (Ménière et al., 2023; Toh et al., 2023).

Second, our findings contribute to the literature on patent system design, patent quality, and examination intensity (Caillaud and Duchêne, 2011; Schuett, 2013a,b; Atal and Bar, 2014; Schankerman and Schuett, 2022; Castaldi et al., 2024; Liu, 2024). Patent quality is multidimensional; citations, claims, and renewals capture aspects of underlying value (Lanjouw and Schankerman, 2004), while we study a patent's ability to attract external finance. Examination design shapes patent outcomes through examiner workloads (Shu et al., 2022), examination standards (Allison and Hunter, 2006; Sampat, 2010; Langinier and Marcoul, 2016), and applicants' decisions about filing, claim scope, and abandonment (Matcham and Schankerman, 2026). Historical evidence on the effects of patent system design comes from Hanlon and Jaworski (2022), who examine how military procurement rights augmented patent protection and affected aircraft innovation. Our study likewise exploits historical institutional change but examines a different component of patent system design: how information produced by examination affects firms' access to external finance.

Third, our findings add to historical research on the institutional foundations of innovation and long-run development (Lerner, 2002; Mokyr, 2009; Moser, 2013; Kelly et al., 2021; Billington et al., 2024; Cinnirella et al., 2025; Hanlon, 2025; Berkes et al., 2026). Highlights of this historical literature include studies of bank failures and innovation during the Great Depression (Nanda and Nicholas, 2014), the public equity performance of patenting firms during the Second Industrial Revolution (Lehmann-Hasemeyer and Streb, 2016), and the effects of French-imposed legal reforms on innovation in nineteenth-century Germany (Donges et al., 2023). Using Britain's 1905 reform, we identify state-produced information as a mechanism through which institutional design affects capital allocation. By making innovation more credible to external investors, patent examination directed additional finance towards innovative firms and enabled their growth.

The paper proceeds as follows. Section 2 describes the institutional setting. Section 3 sets out the analytical framework and empirical predictions. Section 4 introduces the data; Section 5 presents the empirical strategy and results; and Section 6 concludes. The supplementary materials provide qualitative evidence (Appendix A), data construction and supplementary patent statistics (Appendix B), and additional results and robustness exercises (Appendix C).

and reduced interest rates for patenting firms.

2 Institutional setting

2.1 From registration to examination

Patent systems differ in how they assess applications before granting protection (de Saint-Georges and van Pottelsberghe de la Potterie, 2013; Picard and van Pottelsberghe de la Potterie, 2013; Harhoff, 2016). Registration systems check documents and formalities but leave validity largely to subsequent enforcement through the courts. Prior-art examination adds an official search of earlier technical claims, while full substantive examination also empowers officials to determine whether an invention meets standards such as novelty and inventive step. Britain's 1905 reform occupied the intermediate category: it introduced a prior-art search without full substantive examination.

The Patents, Designs, and Trade Marks Act 1883 had previously reshaped the British patent system in two respects (Van Dulken, 1999; Nicholas, 2011a). It introduced a limited formalities examination and reduced the initial cost of a patent from £25 to £4, while introducing renewal fees.[5] Patenting rose sharply, but Nicholas (2011a) finds no corresponding increase in innovation using renewal- and citation-based measures. By the turn of the century, Britain therefore had an accessible patent system with limited formalities review but no systematic public investigation of prior art.

Concern grew over patents for obvious or previously patented inventions and the resulting risk of wasteful litigation. The Board of Trade's 1901 Departmental Committee on the working of the Patents Acts, chaired by Sir Edward Fry, concluded that up to 42 per cent of patents granted between 1898 and 1900 covered inventions that were neither new, novel, nor non-obvious (House of Commons, 1901). Its recommendations shaped the Patents Act 1902, which took effect in January 1905. The Act otherwise left the substantive rights conferred by patents largely unchanged and, apart from the new £1 post-search sealing fee, retained the low-fee structure, renewal schedule, and 14-year term. Its central innovation was to make prior-art information public through amendments to specifications and references to earlier patents.

[Figure 1 about here]

2.2 How the examination process worked

The Patent Office recruited technically trained examiners to compare applications with earlier British patents (see Figure 1). Although the search required additional staff, its administrative

[5] Examiners checked whether the provisional and complete specifications were consistent and required redrafting where they were not.

cost remained modest: salary expenditure rose from 27 per cent of Patent Office income in 1902 to 34 per cent in 1905 and 1906.[6]

Examiners searched the previous 50 years of British patents for claims that might have pre-empted, or 'anticipated', the claimed invention.[7] They reported the results to the applicant and, where necessary, supplied a list of anticipating specifications. Applicants could amend the specification to define its novelty more precisely or withdraw. If an applicant proceeded without adequate revision, the Comptroller-General could append the relevant earlier specifications to the published patent as 'forced references'.

After receiving the search report, applicants wishing to proceed paid a new £1 sealing fee, raising the total cost of a sealed patent from £4 to £5.[8] This sequencing allowed applicants to decide whether to pursue their rights in light of the report. Examination could improve the resulting patent pool through three margins: anticipatory non-entry by applicants expecting adverse findings, amendment in response to the search, and withdrawal before sealing.[9] Because the additional fee was small and payable only after the report, any anticipatory non-entry would arise from the prospect of examination rather than the price of applying.

Where an applicant proceeded without sufficient amendment, the Comptroller-General held a hearing and determined which references to append, subject to appeal to the Law Officer.[10] Neither examiners nor the Comptroller-General possessed a general power to reject applications for insufficient novelty or inventive step.[11]

The patent system was further modified during our sample period by legislation enacted in 1907 and effective from 1 January 1908. Its principal change was to transfer the narrow refusal power from the Law Officer to the Comptroller-General without materially altering the prior-art search or its fee structure.[12] Under the revised procedure, the Comptroller-General could refuse an application only where, after amendment and a hearing, the invention remained 'wholly and specifically claimed' in an earlier specification. Examiners still could not refuse

[6] Authors' calculations using data published in the *Reports of the Comptroller-General* (1902–1906).

[7] Other patent offices conducted broader novelty searches; contemporaries recognised the distinction (see *The Times*, 25 December 1908).

[8] £1 in 1905 is equivalent to approximately £150 in 2025 prices (https://www.measuringworth.com/).

[9] As Appendix Table B1 shows, patentees amended 57.2% of patents sealed in 1905 and 65.2% in 1910 following the search. Including forced references, these shares rise to 60.8% and 67.1%. In the same years, 447 and 615 applications did not proceed to sealing, equivalent to 1.6% and 2.0% of applications filed.

[10] The Law Officer was a senior political office-holder, while the Comptroller-General was the senior civil servant heading the Patent Office.

[11] Only 14 patents were refused outright by the Law Officer following an appeal in 1905, with the annual number remaining between 10 and 14 until 1912 (*Reports of the Comptroller-General*, 1905–1913).

[12] See the Patents Act 1902, s.1(6–7); the Patents and Designs (Amendment) Act 1907, s.7; the Patents and Designs Act 1907, s.7(2–5); and the Patents Act 1949, s.6(2). The holder of the earlier patent was not a party to the examination stage procedure. The 1907 legislation otherwise left the examination regime materially unchanged (Adams, 2007).

applications, and a general power to reject them on substantive patentability grounds was not introduced until the Patents Act 1949. The 1907 legislation therefore did not convert prior-art search into full substantive examination.[13]

2.3 Patents, information, and external finance

The 1905 reform supplemented disparate private signals about the strength of patent claims with a standardised public record of prior-art scrutiny. The Comptroller-General claimed that the search had 'materially enhanced the value of patents granted in the United Kingdom' (House of Commons, 1910, p. 4). It nevertheless provided no guarantee of novelty, legal validity, or commercial success.[14]

Before 1905, firms sometimes commissioned patent agents and technical experts to assess their inventions and reproduced favourable opinions in prospectuses. Because these assessments were issuer-funded and varied across firms, investors had to evaluate both the expert and the patent. Examination instead supplied a common Patent Office record. Contemporary commentary connected that record to the saleable value of patented technology while continuing to caution that examination did not guarantee legal validity; Appendix A provides examples.

Patents also entered contemporary financing arrangements directly. The floating charge securing bondholders covered a company's assets as a whole, so its patent portfolio *de facto* formed part of the security supporting its bonds (Coyle and Turner, 2013). Patenting firms also listed patents among their balance-sheet assets, while prospectuses described their value and sometimes referred explicitly to their use as security for bondholders (see Appendix A).[15]

Examination therefore increased the informational value of patents through both screening of the patent pool and standardised public certification of prior-art scrutiny. Section 3 formalises how these mechanisms could affect investors' beliefs about patent quality and expected value, and thereby firms' access to external finance.

[13] Berkes et al. (2026, p. 2, fn. 4), following Van Dulken (1999, p. 28), incorrectly characterise the 1907 refusal power as full examination. Section 7(4) permitted refusal only where, after amendment and a hearing, the invention remained 'wholly and specifically claimed' in an earlier specification; it did not authorise general rejection for insufficient novelty or inventive step. Treating this limited power as full examination conflates it with the general substantive power introduced in 1949.

[14] Contemporaries anticipated this risk: witnesses to the 1901 Parliamentary Committee worried that the public might place excessive confidence in a favourable examination result; see also contemporary opposition to the proposed legislation in *The Economist* (12 April 1902).

[15] Balance sheets generally combine patents with other intangible assets rather than reporting a separate value, preventing us from analysing this item directly.

3 Analytical framework and empirical predictions

This section presents a parsimonious analytical framework linking patent examination to external finance and firm growth. Its parameters are not estimated structurally; instead, the framework sets out the economic logic and derives predictions for the empirical analysis that follows. The model's screening block adapts Schuett's (2013b) definition of patent quality and Schankerman and Schuett's (2022) distinction between deterrence before application and detection during examination.[16] We apply this screening logic to Britain's 1905 reform and add a financing stage in which state certification informs investors' assessment of expected pledgeable value.[17]

3.1 Environment and timing

Consider a firm with a patentable invention seeking external finance for commercialisation and subsequent R&D. The firm privately observes the joint quality of the invention and its patent claim, represented by the type $\theta \in \{H, L\}$; investors do not. Type H combines a more technically credible invention with a stronger patent claim, while type L combines a less credible invention with a weaker claim. Type captures prospective invention value and the likely effectiveness of legal protection, not commercial success or the firm's overall quality.

Before receiving information from the patent system, investors assign probability $\Pr(\theta = H) = \mu$, where $0 < \mu < 1$. Thus, $\mu = 0.6$ means that they initially expect 60 per cent of invention-patent pairs to be type H. The model's timing separates the invention-patent pair's underlying quality, the information available to investors, and the likely effectiveness of legal protection:

(1) The firm observes the quality of its invention and patent claim and decides whether to enter the patent system.

(2) Before 1905, the Patent Office registers the application without examination. After 1905, it searches prior art, following which the applicant may proceed, amend the specification, or withdraw.

(3) Investors observe whether the resulting patent was registered or examined and update their assessment of the invention-patent pair, including the likely effectiveness of its legal protection. Examination changes the information available to investors but does

[16] In Schankerman and Schuett (2022), types distinguish inventions by whether patent protection is needed for development; ours capture the investor-assessed strength of existing invention-patent pairs. We abstract from examiner incentives, licensing, endogenous court challenges, and welfare analysis.

[17] Schuett (2013a) also models external finance, but examination there relaxes an *ex ante* moral hazard constraint before R&D is funded. Our financing stage follows invention and examination.

not guarantee protection in court.

(4) The firm raises finance through bonds, bank borrowing, or outside equity. The amount available, including the quantity of bonds investors will absorb, determines whether its planned investment can proceed.

3.2 Examination, patent quality, and probabilistic property rights

Patent systems can screen applications through fees, examination, or both (Encaoua et al., 2006; de Rassenfosse and van Pottelsberghe de la Potterie, 2013; Kou et al., 2013). We study prior-art search examination, a standardised state process that scrutinised claims against earlier specifications and produced public information, considering the pre- and post-1905 regimes in turn.

Before 1905. Low fees induced little self-selection, and patents were registered without substantive examination. Registration supplied no common public signal about the quality of the invention-patent pair, leaving types H and L pooled from investors' perspective (Akerlof, 1970). In principle, privately informed firms can distinguish themselves through costly observable signals (Spence, 1973; Leland and Pyle, 1977). Firms commissioned patent agents, counsel, and other experts, but these issuer-funded, bespoke opinions were imperfect substitutes for a common public signal (see Section 2 and Appendix A). We abstract from these signals, so observing a registered patent leaves investors' initial assessment unchanged:

$$\mu_0 \equiv \Pr(\theta = H \mid \text{registered patent}) = \mu. \tag{1}$$

After 1905. The state searched prior art and reported its findings to applicants. Applicants could amend their specification, withdraw, or proceed without amendment, in which case the Patent Office could force references to pre-empting specifications.[18] The prospect of examination might also deter applicants expecting adverse findings, allowing selection before any search. The new £1 sealing fee, payable only after the search was completed, raised total fees from £4 to £5 but did not materially strengthen fee-based screening. Remaining patents carried standardised state certification that their claims had undergone prior-art scrutiny, whether or not investors inspected the underlying record.

Let $e \in [0, 1]$ measure the examination-induced reduction in the odds of type L relative to type H in the post-reform patent pool. This reduction may arise through anticipatory

[18] Forced references could strengthen the informational effect of examination without changing the composition of the patent pool. We do not model this channel separately because forced references were rare, and none of the patents held by firms in our sample contained one.

non-entry, withdrawal following the search, or quality-improving amendment. For example, $e = 0.5$ means that the odds of type L relative to type H are 50 per cent lower in the examined than in the registered patent pool. Following Schuett (2013b), let μ_1 denote the posterior probability that a post-1905 patent is type H, an index of average patent quality:

$$\mu_1 \equiv \Pr(\theta = H \mid \text{examined patent}) = \frac{\mu}{\mu + (1-\mu)(1-e)}. \tag{2}$$

Continuing the example, if $\mu = 0.6$ and $e = 0.5$, then $\mu_1 = 0.75$. Investors assign an examined patent a 75 per cent probability of being type H, compared with 60 per cent for a registered patent. We do not interpret this expression as an application-level account of examination. Rather, it captures the combined effect of the three routes on the composition of the patent pool.

Proposition 1: Examination raises average patent quality

If examination or its prospect reduces the share of type L in the patent pool through non-entry, withdrawal, or amendment, then $e > 0$ and $\mu_1 > \mu_0$. Average invention-patent quality is therefore higher after the reform.

Average patent quality matters because investors care about both the underlying invention and its legal protection. Let $X_\theta \geq 0$ denote the pledgeable cash flow expected from an invention of type θ without effective exclusion, where $X_H \geq X_L$. Let r_θ denote the probability that its patent can be protected if challenged, with $0 \leq r_L < r_H \leq 1$, and let $\Delta > 0$ denote the additional pledgeable value created by effective protection. Expected pledgeable value is $V_\theta \equiv X_\theta + r_\theta \Delta$, so $V_H > V_L$.

The search did not certify commercial success, but evidence of technical distinctiveness could make prospective cash flows more credible. Differences in X_θ capture this information, while $r_\theta \Delta$ captures the patent as a probabilistic property right (following Lemley and Shapiro, 2005). Together they determine expected pledgeable value.[19]

In regime j, expected pledgeable value is $\mu_j V_H + (1-\mu_j)V_L$. Since $V_H > V_L$, the increase from μ_0 to μ_1 raises the average pledgeable value of patents. This can reflect more credible prospective cash flows, a higher average probability of effective protection, or both. Our empirical analysis cannot distinguish these components, but both ultimately reflect information generated by examination rather than a change in the legal standard applied by the courts.[20]

[19]The result also holds in the special case $X_H = X_L$. Examination then raises expected pledgeable value solely by increasing the average probability that a patent can be protected.

[20]No digitised database of patent infringement cases is available for this historical period. Constructing one

3.3 External finance and firm growth

The financing problem. A firm has a profitable investment opportunity requiring expenditure I and internal funds $W < I$, leaving an external financing requirement $F = I - W > 0$. It can raise this amount through bonds, bank loans, or outside equity. Let B, L, and E denote the cash raised from each source. Examination may alter the amount available from each, while relative financing costs influence the mix the firm chooses. Any such mix must satisfy the funding identity:

$$F = B + L + E. \tag{3}$$

Bond capacity. Bonds provide a tractable case. Let $\overline{B}_j$ denote the maximum bond finance investors will provide at prevailing terms in regime j. It measures market demand rather than the firm's legal authority to issue bonds or its actual bond debt. We hold other creditors' claims fixed and treat V_θ as the pledgeable value available to bondholders after any prior claims. If $R_B > 1$ is the gross return required by bondholders, bond capacity is:

$$\overline{B}_j = \frac{1}{R_B}\left[\mu_j V_H + (1-\mu_j)V_L\right], \qquad j \in \{0,1\}. \tag{4}$$

The term in square brackets is expected pledgeable value described in the preceding subsection; division by R_B converts future cash flow into the amount investors will provide today. This accords with evidence that stronger property rights and more pledgeable intangible assets increase debt capacity and creditor recovery (Campello and Giambona, 2013; Hochberg et al., 2018; Mann, 2018; Suh, 2023; Luck and Santos, 2024).[21]

When examination improves the patent pool, investors place more weight on the pledgeable value of a type H patent and less on that of type L. Holding R_B and other creditors' claims fixed, bond capacity rises by:

$$\overline{B}_1 - \overline{B}_0 = \frac{(\mu_1 - \mu_0)(V_H - V_L)}{R_B} > 0. \tag{5}$$

The first difference is the increase in investors' assessed probability that a patent is type H; the second is the extra pledgeable value of that type. In the example above, $\mu_0 = 0.6$ and $\mu_1 = 0.75$, so bond capacity rises by $0.15(V_H - V_L)/R_B$. Actual bond finance satisfies $B \leq \overline{B}_j$, so higher capacity need not lead to additional bond issuance.

from Patent Office records would be necessary to test the legal-protection channel directly.

[21] Chen et al. (2026) show that a Chinese legal reform increasing asset pledgeability expanded debt capacity and innovation, especially where enforcement was stronger.

Proposition 2: The bond capacity effect

If examination raises average patent quality, so that $\mu_1 > \mu_0$, it increases the amount of bond finance available to patenting firms at prevailing terms.

Prediction 1: Bond capacity

Following the introduction of examination, patenting firms should be able to place more bonds at prevailing coupons and other contractual terms, including bonds already authorised but previously unsupported by sufficient demand.

Financing mix. Whether greater bond capacity changes firms' financing mix depends on the alternatives. Bond markets pooled dispersed savings, giving firms access to a larger funding base than a single relationship bank, but their investors could not cheaply investigate borrowers themselves. Relationship banks could acquire private information through screening and monitoring. Examination therefore plausibly added more information for bond investors than for banks, although banks might also benefit where their private information was incomplete (Leland and Pyle, 1977; Saidi and Žaldokas, 2021).[22]

Outside equity presents a further possibility. Under registration, investors may discount new shares because they cannot assess the value of a firm's inventions and patent claims. This problem is acute for innovative firms with intangible assets and uncertain returns (Myers and Majluf, 1984; Harhoff, 2009; Fulghieri et al., 2020). If certification helps investors value the issuing firm, it can reduce that discount, allowing a given ownership stake to raise more cash or a given amount of cash to be raised with less dilution (Korajczyk et al., 1991). For example, a stake worth £1 to existing shareholders might raise 90p under registration but 95p following examination. Historical evidence linking patents to equity market valuations makes this response plausible (Lehmann-Hasemeyer and Streb, 2016). It does not follow from $\mu_1 > \mu_0$ alone, however, because that result concerns average patent quality rather than the value of the issuing firm.

Whether firms use greater bond capacity or issue equity depends on relative financing costs. Pecking order theory favours internal finance and debt when shares risk undervaluation (Myers, 1984).[23] Equity can also dilute existing owners' control rights (Jensen and Meckling, 1976). But certification may make equity more attractive if it sufficiently reduces the discount on new shares. Bank loans involve intermediation costs, while bonds can be costly in distress

[22] Britain's developed corporate debt markets made bond financing a plausible margin of response to state certification (Coyle and Turner, 2013; Coyle et al., 2019).

[23] Trade-off theory also predicts a tax-driven preference for debt (Modigliani and Miller, 1963; Kraus and Litzenberger, 1973), but no corporate tax system existed in our period (Braggion and Moore, 2011).

(Bolton and Freixas, 2000). Examination raises bond capacity and may improve access to bank loans or equity, but does not determine firms' financing mix.

Prediction 2: Bonds, banks, and equity

If public certification is more valuable to dispersed bond investors than to relationship banks, patenting firms should increase their reliance on bond debt relative to bank borrowing. The response of outside equity remains open.

Firm growth. A profitable project requires investment I, towards which the firm can contribute $W < I$ from internal funds. Let $\overline{F}_j$ denote the maximum external finance available from bonds, bank loans, and outside equity together in regime j, taking their terms and competing claims into account. For the same project and internal funds, examination makes investment feasible when:

$$W + \overline{F}_0 < I \leq W + \overline{F}_1. \tag{6}$$

Proposition 3: Financing capacity and firm growth

If examination expands overall financing capacity enough for a profitable project to satisfy Equation (6), the firm can undertake it. Investment rises and, insofar as it expands the firm's asset base, so does measured firm size.

Greater bond capacity can raise $\overline{F}_j$ where market demand for a firm's bonds had constrained the finance it could obtain, including where bonds had already been authorised but could not be placed. Improved access to bank loans or better terms for outside equity provide other possible routes. Firms whose projects were already financeable, or remain unfinanceable after examination, need not change investment. Bank debt and equity may also rise in absolute terms among firms that expand.

Prediction 3: Financing capacity and growth

Where examination relaxes a binding overall financing constraint, the total assets of patenting firms should increase.

Taken together, the model shows that examination raises bond capacity at prevailing terms when it improves average patent quality. Bank loans and equity may respond as well, leaving the composition of finance raised by firms open. Where the resulting increase in overall financing capacity enables profitable investment, the firm's asset base grows.

4 Data and descriptive statistics

To examine how prior-art search examination affected patenting firms' ability to raise external finance, we focus on stock exchange-listed companies. This provides reliable financial data and restricts attention to firms with access to both banks and capital markets. It also complements a literature focused predominantly on innovation and finance in unlisted startups. We construct a dataset of firms listed on the London Stock Exchange during 1899–1913, comprising all identifiable firms that patented before or after the introduction of examination and a randomly selected control group of firms that never patented.

We identify patenting firms using the European Patent Office's PATSTAT Biblio database (Autumn 2016 version). Although PATSTAT's historical coverage is incomplete, its British patent data are comprehensive from the late 1890s onwards (Billington and Hanna, 2021). We therefore begin in 1899, providing a substantial pre-reform window, and end in 1913 to avoid the disruption caused by World War I. We manually match patentees to companies listed in the *Investor's Monthly Manual* (IMM) and *Stock Exchange Daily Official Lists* (SEDOL). The sample is restricted to the 'Commercial, Industrial' and 'Iron, Coal, Steel' categories, which contain the overwhelming majority of listed British patenting companies during this period. The control firms are randomly drawn from the same categories.

We then hand-collect annual balance sheets for each firm. The final dataset contains 122 patenting companies and 173 non-patenting controls, yielding 2,509 firm-year observations. Of these firms, 17 patented only before 1905, 61 only from 1905 onwards, 44 in both periods, and 173 in neither. Thus, 61 firms patented before 1905 and 105 from 1905 onwards, with the latter constituting the treated group. Appendix B describes the sample construction, matching procedure, and data sources in detail; Appendix Table B4 defines the variables.

4.1 Patent data

Figure 2 plots sealed patents relative to applications, complete specifications, and provisional specifications around the reform. Sealed patents closely track complete specifications before 1905 but diverge from them afterwards, consistent with post-search abandonment. They also rise relative to provisional specifications, consistent with examination discouraging non-novel applications. Appendix Table B1 provides complementary evidence: only around one-third of complete specifications were wholly unanticipated, most sealed patents were amended after search, and both whole anticipation and forced references declined over time.

[Figure 2 about here]

Appendix Tables B2 and B3 provide complementary PATSTAT statistics for British patentees and the listed firms in our sample. Table B2 compares patent stocks, references, abandonment, and forward citations before and after the reform, while Table B3 reports annual patenting and citation activity among the listed firms. Abandonment increased following the reform, and applications filed by listed firms subsequently received more forward citations on average. Viewed alongside Figure 2, these patterns are consistent with examination improving the average quality and informational value of the patent pool.

4.2 Financial data

Table 2 reports financial summary statistics for patenting and non-patenting firms before and after the reform. Our main dependent variable is total debt divided by total assets. Before examination, patenting and non-patenting firms had similar average debt-to-assets ratios of 0.25 and 0.24. Afterwards, the ratio increased to 0.28 among patenting firms but remained unchanged among controls.

The descriptive divergence is concentrated in bonds. Before 1905, patenting and non-patenting firms had average bond debt-to-assets ratios of 0.16 and 0.15. After 1905, the ratio increased to 0.18 among patenting firms but remained unchanged among controls. Bank borrowing remained stable, while trade credit increased similarly in both groups. These patterns anticipate the paper's principal result that examination expanded patenting firms' access to arm's-length bond finance rather than relationship-based or transactional credit.

[Table 2 about here]

The similarity in pre-reform leverage reflects the institutional setting rather than an absence of financing frictions. Britain possessed highly developed corporate debt markets, listed firms generally had access to debt finance, and creditor rights were strong, particularly for bondholders (Hannah, 2007; Coyle and Turner, 2013; Hannah, 2015; Coyle et al., 2019). This environment provided scope for more credible patent information to increase debt capacity (Mann, 2018). The relevant margin is therefore not initial access to debt, but whether examination enabled patenting firms to raise additional arm's-length finance by increasing the credibility and expected pledgeable value of their patents.

Table 3 reports descriptive statistics for the pooled firm-year sample. Patenting and non-patenting firms are broadly similar in profitability but differ in some observable characteristics. Patenting firms tend to be larger and are slightly more likely to be family or director controlled, whereas non-patenting firms are somewhat more likely to be headquartered in London. The

empirical analysis accounts for these differences and tests whether the descriptive financing patterns survive firm and year fixed effects and other robustness exercises.

[Table 3 about here]

5 Empirical strategy and results

5.1 Difference-in-means descriptive statistics

We treat the 1905 introduction of patent examination in Britain as an exogenous policy intervention in a difference-in-differences framework. We compare patenting with non-patenting firms. First, we use simple difference-in-means tests to examine whether differences in our key debt variables are statistically significant. Table 4 reports average debt financing for patenting and non-patenting firms before and after the introduction of prior-art search examination. We run tests using the year of patent grant (Panel A) and patent application (Panel B) to identify patenting firms. These results confirm the average trends identified in Section 4; patenting firms borrow significantly more relative to their assets after the introduction of examination, whereas there is no significant change in non-patenting firm debt.

[Table 4 about here]

Panel A shows that the average total debt-to-total assets ratio for firms with granted patents increases by a statistically significant 11.44 per cent after patent examination is introduced. The corresponding 2.78 per cent increase among non-patenting firms is statistically insignificant. Before examination, firms with patents borrowed 4.59 per cent more than non-patenting firms relative to assets (24.83% versus 23.74%). After the introduction of examination, average debt ratios for firms with examined patents were 13.36 per cent higher than those for non-patenting firms (27.67% versus 24.41%).

We split firms' total debt into capital market debt (i.e., bond debt), bank debt and trade credit. Patenting firms show statistically significant increases in both average 'bond' and 'trade' debt ratios after examination. Patenting firms' 'bond' debt ratios increased by 10.88 per cent (16.09% versus 17.84%) after examination, while there is no significant change in bond financing for non-patenting firms. After examination, patenting firms borrow 17.99 per cent more than non-patenting firms through bond markets (15.12% versus 17.84%). While non-patenting firms increase their bank borrowing after 1905, there is no statistically significant change in average bank borrowing by patenting firms.

Panel B reports difference-in-means tests using the year of patent application to identify when firms are treated. The changes before and after examination are similar to those in Panel A, but larger. Comparing means between Panels A and B, the data show that before examination is introduced, a firm that has filed a patent application borrows at the same level as non-patenting firms on average (0.2399 versus 0.2390), and less than firms with a granted patent (0.2399 versus 0.2483). After the introduction of examination, firms in the patent application group borrow, on average, at an almost identical level to those with granted patents (0.2765 versus 0.2767). This change suggests that even as-yet unexamined patent filings may have an additional signalling effect in the new examination regime and implies an expectation, on the part of lenders, of a higher-quality patent application pool.[24] These data are consistent with the hypothesis that prior-art search examination increased the credibility of patent-related information, thereby improving the value of patents to external investors.

5.2 Total debt ratio difference-in-differences regression results

Next, we introduce our more formal difference-in-differences analysis to examine the relationship between patent examination and firm financing more robustly. Our analysis is based on the following two-way fixed effects difference-in-differences model:

$$\begin{aligned} Y_{it} = {} & \alpha\,\text{Examination}_t + \beta\,\text{PatentingFirm}_{it} \\ & + \gamma(\text{Examination}_t \times \text{PatentingFirm}_{it}) + \theta X_{it} + \delta_i + \tau_t + \varepsilon_{it}. \end{aligned} \tag{7}$$

where Y_{it} is the total debt-to-total assets ratio; Examination_t equals one during the examination regime (1905–1913) and zero otherwise; and $\text{PatentingFirm}_{it}$ equals one once a firm has obtained a patent under the relevant regime. The patenting indicator is regime-specific and resets in 1905: a firm that patented only before 1905 is not treated after the reform unless and until it subsequently receives an examined patent. In this specification, γ is the difference-in-differences estimate and captures the additional change in financing among patenting firms following the introduction of examination. The total post-reform difference between patenting and non-patenting firms is given by $\beta + \gamma$. Because patenting remains a firm's choice, our identification combines the exogenous timing of the examination reform with within-firm variation and comparisons between patenting and non-patenting firms. The identifying assumption is that, absent patent examination, the debt outcomes of treated and comparison firms would have followed parallel trends. We assess this assumption using event-study estimates and formal

[24] With the caveat that there is an overlap between the 'filing' and 'grant' groups because 22% of our firms had their patent filed and granted in the same year.

pre-treatment tests after presenting the baseline results (see Table C1).

The vector X_{it} contains firm-level controls for profitability and growth opportunities (ROA, ROE, and Tobin's Q), self-financing capacity (Reserves and Retained Earnings/Assets), size (Log Total Share Book Value), maturity (Retained Earnings/Ordinary Equity), and age. Firm fixed effects, δ_i, control for time-invariant heterogeneity, while year fixed effects, τ_t, absorb common time-varying factors; ε_{it} is the error term.[25] The subscripts i and t index firms and years, respectively. Parameter γ captures the relationship between patent examination and patenting firms' financing decisions relative to non-patenting firms. We report specifications using conventional and firm-clustered standard errors.

[Table 5 about here]

Table 5 reports results from the baseline difference-in-differences panel regressions on the total debt-to-total assets ratio of all firms, from 1899–1913. Our interaction term in column 1 ('Patenting x Examination') shows that firms patenting after 1905 have a 0.022 higher total debt-to-total assets ratio. In other words, firms that have an examined patent borrow significantly more compared to the pre-reform period and firms that did not patent. Columns 2-8 report our findings are consistent and robust to controlling for various firm characteristics that may also explain differences in borrowing. The estimated effect of an examined patent increases to between 0.024 and 0.027, and is significant at both the 5 and 1 per cent level. Compared to a pre-reform average total debt-to-total assets ratio of 0.25 for patenting firms, this represents a relative increase in leverage of between 9.6 and 10.8 per cent following the grant of an examined patent.

Consistent with what we would expect, columns 2 to 8 suggest that more profitable firms borrow less, firms with greater reserves and retained earnings relative to their size borrow less, and larger firms borrow less. Firm age does not have a significant effect on borrowing, all else constant. Column 4 shows that firms patenting in 1903 or 1904 – that is, after the new examination regime is announced in the 1902 Act, but before it is implemented – do not borrow more. This suggests that the result is driven by the institutional change of prior-art search patent examination itself, rather than anticipation of a new examination regime. This finding helps rule out alternative explanations that would suggest that results are merely capturing a pre-existing trend or a reaction by firms to the news of the impending legislation.

[25]Table B4 provides full variable definitions and historical sources.

5.3 Debt breakdown difference-in-differences regression results

Firms in our study window accessed debt through capital markets (which tends to be longer maturity credit), bank lending (which is relationship-based and often shorter term), or using trade credit (which is short-term business-to-business borrowing). To investigate whether the average differences in source of credit observed in Sections 5.1 and 5.2 are robust, we analyse bond debt (debentures raised on capital markets) and non-bond debt (other debt) separately.

[Table 6 about here]

[Table 7 about here]

[Table 8 about here]

Tables 6 and 7 report difference-in-differences estimates for bond debt and other non-bond debt, respectively, over 1899–1913. The positive interaction term in Table 6 indicates that firms patenting under the examination regime increased their bond debt-to-assets ratios by between 0.029 and 0.031 relative to non-patenting firms and the pre-reform difference between the groups. Compared with the pre-reform average of 0.1609 for listed patenting firms (Table 4), this represents an increase in capital market borrowing of between 18.0 and 19.3 per cent. Table 7 shows no comparable examination effect for other non-bond debt.

Appendix Figure C1 examines the timing of these effects and the identifying parallel-trends assumption using event-study extensions of our baseline specifications. Coefficients are measured relative to the year immediately before a firm receives its first examined patent. In both panels, the pre-treatment estimates are statistically indistinguishable from zero, providing no evidence of a systematic differential trend in either total debt or bond debt before treatment. Post-treatment point estimates are generally positive and become larger over time, although several annual estimates are imprecisely estimated. The largest effects emerge several years after the first examined patent, particularly for bond debt. This timing is consistent with examination expanding firms' borrowing capacity after the patent record becomes public, rather than capturing a pre-existing increase in leverage. Formal tests of the pre-reform period reported in Table C1 similarly find no statistically significant difference between the trends of patenting and non-patenting firms. The results are consistent whether treatment is dated using patent applications or grants.[26]

Table 8 splits 'other borrowing' further into 'bank borrowing' and 'trade credit'. The data show that there is no significant change in trade credit associated with companies which have

[26]The Callaway and Sant'Anna (2021) estimates discussed in Section 5.4 and reported in Table C7 also produce small and statistically insignificant pre-treatment estimates.

examined patents (Panel B), however there is a negative association with examined patents and bank borrowing (Panel A). This apparent negative association with bank borrowing is driven by a divergence between patenting and non-patenting firms. Non-patenting firms borrow relatively more from banks and patenting firms relatively less, in the period following the introduction of examination (see Table 4). However, the levels of bank borrowing are very small, at around 3 per cent of total borrowing for patenting firms, and between 3 and 5 per cent for non-patenting firms.[27]

Taken together, the debt results are consistent with Prediction 2: bond borrowing rose without a comparable increase in bank borrowing. Examined patents can act as a signalling device through which the underlying value of an innovative firm is revealed to wider market participants. By making patent-related information more observable and credible, prior-art search examination may have reduced information asymmetries and enabled greater access to capital market debt. The concentration of the financing effect in bond markets is consistent with the information channel being particularly important: arm's-length investors were the capital providers most likely to benefit from comparable, publicly observable information about anticipation, claim scope, and novelty. The value of examined patents to external investors therefore reflected both their legal rights and the public information generated by examination.

We propose two explanations for the observed negative association between firms in possession of examined patents and their rate of bank borrowing. First, it is possible that relationship-based banking may mean that banks were already more aware than other market participants of the innovative prospects of firms, which limited their capacity for further bank borrowing. For capital market creditors, patent examination may effectively act as a substitute to the relationship-based private information banks enjoy (Saidi and Žaldokas, 2021). Additionally, innovative firms may prefer capital market debt, which is typically more arm's-length, of larger scale, and longer term, particularly given evidence that traditional, risk-averse bank lending is often ill-suited for financing inherently uncertain and intangible innovation (Cornaggia et al., 2015). We suggest that examined patents enabled firms to more easily pursue capital market debt financing options, which they preferred to using banks. This interpretation is consistent with evidence that relationship lenders rely more heavily on private information when valuing collateral, whereas arm's-length creditors depend more on publicly verifiable signals (Luck and Santos, 2024), making patent examination particularly valuable for accessing capital market debt.

[27]This is consistent with studies showing that banks were not a particularly important source of finance for firms in this period (Collins and Baker, 2001, 2003).

[Table 9 about here]

Table 9, column 1, examines whether having an examined patent is associated with the cost of capital market debt. The dependent variable is the weighted average coupon on a firm's issued bonds. We find no significant association between examined patents and borrowing costs. Combined with the increase in bond quantities reported in Table 6, this result is consistent with Prediction 1: examination increased the market's capacity to absorb firms' bonds without requiring a reduction in coupons. Debenture holders in the UK during this period very rarely lost money through default. Debentures were secured by all present and future firm assets, and their holders ranked ahead of other creditors, meaning that firms with successful debt issues tended to borrow at similar rates (Coyle and Turner, 2013). Columns 2 and 3 show lower ratios of common and preference equity to assets among firms with examined patents. Our model does not predict a clear equity response because certification may make equity easier to raise even as firms make greater use of debt. The estimates show that the financing response was concentrated in bonds, although the lower equity ratios do not imply that equity issuance fell in absolute terms.

The positive association between examined patents and debt-to-assets ratios, alongside the negative association with equity-to-assets ratios, could reflect either a change in capital structure or an expansion in firms' overall financing capacity. Prediction 3 distinguishes between these possibilities: if examination enabled firms to finance previously constrained investment, examined patents should be associated with subsequent growth in total assets. Table 10 tests this prediction using the log of total assets as the dependent variable. Columns 2 to 5 control for firm characteristics, including total issued equity in column 5. We find no significant association between the examination period and firm size, and a weak negative association between patenting and firm size. The interaction term is positive and significant, indicating that examined patents are associated with larger firms after the reform. This result is consistent with Prediction 3 and suggests that additional debt supported firm growth rather than simply rebalancing capital structures.

[Table 10 about here]

5.4 Further robustness exercises

To further test the robustness of our results, we examine several alternative specifications, all reported in Tables C2–C8 in the Appendix. First, Table C2 reports pooled regressions that add controls for family ownership, London headquarters, and the two broad industry

categories used by the London Stock Exchange. Family control and London location have no consistent association with debt-to-assets ratios, while firms in Iron, Coal and Steel borrow significantly more relative to assets. Controlling for these characteristics leaves our principal results unchanged: firms with examined patents borrow significantly more, with the increase concentrated in capital market debt, consistent with Prediction 1. However, the London Stock Exchange classification divides our sample into only 'Commercial & Industrial' and 'Iron, Coal and Steel', and therefore cannot absorb more finely grained industry-specific developments.

Second, we address this concern using a more granular 18-category industry classification described in Table B4. We first allow each industry to follow its own linear trend and then include industry-by-year fixed effects, which absorb shocks common to firms within the same industry and year. We also exclude firms in 'Electricity and Power', a patent-intensive sector that experienced substantial expansion during the period. Table C3 shows that the estimated effect on bond debt remains stable, ranging from 0.026 to 0.027, and statistically significant across all three specifications. The total-debt estimates remain positive, although they become smaller and less precise when industry-specific trends or industry-by-year fixed effects are included. Excluding 'Electricity and Power' produces a total-debt estimate of 0.026, significant at the 5 per cent level. These results indicate that the increase in bond finance was not driven by general financing developments within patent-intensive industries.

Third, we examine whether our results are sensitive to the characteristics of the randomly selected control group. Table C4 reports four alternative specifications. Column 1 uses propensity-score matching (PSM) to construct a sample of observably similar firms. We estimate propensity scores using a logistic regression based on firm-level averages of pre-reform profitability (ROA), self-financing ability (Total Reserves/Total Assets), and size (the natural logarithm of Issued Share Capital). We then apply one-to-one nearest-neighbour matching without replacement. Column 2 uses entropy balancing to reweight the control group so that the distribution of the same pre-treatment characteristics matches that of the treated group. Unlike matching, entropy balancing retains the control observations but assigns weights chosen to balance the covariate distributions across the two groups. Column 3 restricts the control group to firms that never patented in either the pre- or post-reform period, thereby excluding firms whose patenting status changed around the introduction of examination. Column 4 restricts the sample to firms observed both before and after the introduction of examination, addressing concerns that the results may be driven by firms appearing exclusively before or after the reform. The results closely match the principal estimates: across the four specifications, the interaction coefficient ranges from 0.022 to 0.026 and remains statistically significant at

least at the 10 per cent level.

Fourth, we test alternative definitions of profitability and treatment timing. Table C5 replaces ROA with Return on Equity (ROE), calculated as net profit divided by the book value of issued share capital. This measure reflects returns to equity holders and, following Aldous et al. (2023), reduces potential measurement error arising from inaccuracies in reported assets. Table C6 identifies treatment using the patent application date rather than the grant date. This provides a stricter definition of the regime change and accounts for patents filed before January 1905 but granted under the new regime. Both exercises produce effects of a similar magnitude, direction, and significance to the principal estimates.

A growing literature shows that two-way fixed-effects (TWFE) difference-in-differences models can yield biased estimates under staggered treatment. Although patent examination was introduced nationally in 1905, firms first received examined patents in different years. We therefore use the estimator developed by Callaway and Sant'Anna (2021), which allows treatment effects to vary across cohorts and time by estimating and aggregating group-time average treatment effects using valid comparison groups.

Table C7 reports the results. In Panel A, the primary estimate is 0.023, implying that receipt of an examined patent increases the total debt-to-assets ratio by approximately 2.3 percentage points. It is statistically significant at the 5 per cent level and close to our principal estimate of 0.024 in Table 5. Using only never-treated firms as comparisons produces the same estimate of 0.023, while restricting the dynamic analysis to treatment cohorts with sufficient support through five years after treatment yields an average effect of 0.022. The similarity suggests that the result is not driven by not-yet-treated comparisons or changing cohort composition, despite the smaller number of cohorts contributing to the balanced estimate.

Panel B reports the event-time estimates. The pre-treatment effects are small, vary in sign, and are individually insignificant, providing no evidence of a systematic differential trend. Post-treatment effects are positive at every horizon and largest at four and five years. In the primary specification, the treatment-year effect is significant at the 10 per cent level, while the four- and five-year effects are significant at the 5 per cent level using pointwise inference. At five years, the estimated effect is 0.053, or 5.3 percentage points, and remains significant using the more conservative simultaneous confidence band. The balanced estimates display the same pattern despite fewer contributing cohorts and wider confidence intervals.

Panel C provides evidence that the estimated financing effect increases with longer post-treatment horizons. In the primary specification, the average effect increases from 0.011 over event times 0 to +3 to 0.039 at +4 and +5. The difference is 0.029 and is statistically

significant at the 5 per cent level. The balanced specification produces very similar estimates: the corresponding averages are 0.012 and 0.041, with a statistically significant difference of 0.028. The similarity across specifications indicates that the increase in leverage at longer horizons is unlikely to be driven by changes in the composition of contributing cohorts.

Finally, we conduct 1,000 placebo simulations with randomised treatment assignment to assess whether the estimated effect could arise from noise or model structure. We randomly assign firms to patenting status and patenting years, constraining the assignment to match the observed proportion of patenting firms (41.4 per cent), and re-estimate the baseline specification for each simulated dataset. Table C8 and Appendix Figure C2 show that the resulting coefficients are centred around zero. Only 6.6 per cent are statistically significant, while just 13 of 1,000 simulations, or 1.3 per cent, produce effects that are both significant and positive. The estimated effect is therefore unlikely to be driven by chance.

Taken together, the matching and entropy balancing exercises, estimates using only never-treated controls, staggered treatment analysis, and placebo exercise support the robustness of the borrowing result. The event-study estimates show no systematic directional trend before treatment. Separately, results by debt type show an increase concentrated in bonds, with no comparable rise in bank borrowing or trade credit. While we cannot rule out selection associated with unobserved changes in firms' prospects coinciding with post-1905 patenting, such selection is unlikely to explain this combination of stability, timing, and financing outcomes.

6 Conclusion

This paper estimates the value of patent examination along a key margin that has received relatively little attention in the economics literature on patent-system design: firms' access to external finance. We exploit Britain's introduction of prior-art search examination in 1905, which moved the patent system from registration towards a limited form of state certification. We find that patenting firms increased their borrowing following the reform, with the increase concentrated in capital market debt.

Our model interprets this result through the public information and state certification produced by examination. The prospect and process of prior-art scrutiny can improve the average quality of the resulting patent pool through anticipatory non-entry, post-search withdrawal, and amendment. Sealed patents consequently carry standardised state certification that their claims underwent prior-art scrutiny. This certification allows investors to update their assessment of the invention's prospective value and the probability that its patent can

be protected if challenged. These components jointly raise expected pledgeable value and bond capacity. Both are informational effects, but we do not observe pledgeable value directly or distinguish them empirically. Certification may also reduce the information discount on new equity, leaving the relative use of bonds and equity open. We therefore examine firms' financing choices and growth without assuming which source of finance will respond most.

The empirical results are consistent with the model's financing mechanism. We find no comparable increase in bank lending or trade credit, and bond coupon rates did not decline. The increase in bond borrowing without a corresponding change in coupons is consistent with an expansion in the market's capacity to absorb firms' bonds at prevailing terms. The concentration of the response in bond markets is also consistent with standardised public information being especially valuable to dispersed, arm's-length investors who lacked the private "soft" information available to relationship banks. Patenting firms reduced their reliance on equity relative to assets. This shift towards debt is our finding, not an unconditional prediction of the model. Examination raises bond capacity, but a sufficiently large reduction in the information discount on new equity could make equity preferable at the margin. The observed bond response is consistent with pecking-order considerations, without implying that bank borrowing or equity fell in absolute terms.

The results are consistent with an increase in firms' total financing capacity. Firms with examined patents subsequently grew, consistent with the additional bond finance supporting the commercialisation and further development of their inventions. Greater borrowing may also have exposed firms to additional financial risk if patented inventions failed to generate their anticipated returns. Future research could examine whether this expansion produced sustainable long-run performance or greater financial fragility. Evidence from patent litigation could also help assess the relative importance of the invention-information and legal-protection channels represented in the model.

These findings remain relevant today because patent examination is widespread but institutionally diverse. As Table 1 shows, contemporary patent routes range from formalities-only registration to prior-art search and full substantive examination. These differences matter because search and substantive examination produce different forms of state certification and involve different costs and delays. Britain's reform sequence allows these components to be examined separately. The 1905 reform introduced prior-art search and a public examination record without giving examiners a general power to reject applications. Our estimates therefore provide evidence on the financing consequences of a low-cost search-based certification separately from a high-cost full substantive examination. They show that even relatively

low-intensity and low-cost screening can generate economically useful public information.

The broader substantive powers introduced under the Patents Act 1949, which took effect in 1950, provide a natural setting for extending this analysis. Studying that reform could help identify the incremental financing and innovation effects of full substantive examination. Contemporary initiatives also demonstrate the continuing policy interest in improving information about intellectual property. Singapore, for example, has piloted a scheme intended to improve disclosure about companies' intellectual property portfolios (Toh et al., 2023). Our evidence suggests that the design of patent examination itself should form part of this policy discussion.

The financing response reflects the interaction between patent reform and early twentieth-century Britain's financial structure. London's corporate debt market offered a setting in which the public information produced by examination could matter to dispersed bond investors. The concentration of the response in bonds shows how financial institutions condition the effects of state certification. The market's subsequent contraction reduced access for firms and investors (Coyle and Turner, 2013). Future research could ask whether examination retained this financing function as bond markets became less accessible, or whether its effects instead operated through bank lending and other sources of external finance. This question has wider relevance wherever outside investors must evaluate uncertain intangible assets using public information shaped by patent-system design.

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TABLES AND FIGURES

Table 1. Patent examination in the United Kingdom and selected present-day systems

Patent route	Formalities examination	Official prior-art search	Full substantive examination
United Kingdom over time			
United Kingdom, 1884–1904	Yes	No	No
United Kingdom, 1905–1949	Yes	Yes	No
United Kingdom, 1950–present	Yes	Yes	Yes
Present-day European systems			
Malta	Yes	No	No
Slovenia	Yes	No	Optional
Luxembourg	Yes	Optional	No
Switzerland	Yes	Optional	No
Ireland	Yes	Optional	Optional
Belgium	Yes	Yes	No
Netherlands	Yes	Yes	No
France	Yes	Yes	Yes
Germany	Yes	Yes	Yes
Italy	Yes	Yes	Yes
European Patent Office	Yes	Yes	Yes
Other present-day systems			
South Africa	Yes	No	No
Brazil	Yes	Yes	Yes
China	Yes	Yes	Yes
India	Yes	Yes	Yes
Japan	Yes	Yes	Yes
Russia	Yes	Yes	Yes
Singapore	Yes	Yes	Yes
United States	Yes	Yes	Yes

Notes: 'Formalities examination' checks procedural requirements. 'Official prior-art search' identifies relevant earlier publications. 'Full substantive examination' assesses patentability and can prevent grant. 'Optional' means that the procedure is not required before every grant. The classifications identify the formal structure of examination rather than its quality. Even among systems requiring substantive examination, offices differ in the stringency of patentability standards, examiner resources and workloads, and the consistency with which decisions are made (Picard and van Pottelsberghe de la Potterie, 2013; de Rassenfosse et al., 2021).

United Kingdom over time: The periods follow the commencement of the 1883 Act on 1 January 1884, prior-art search examination on 1 January 1905, and the 1949 Act on 1 January 1950. The 1905 regime produced a public search record but did not give examiners a general substantive power to refuse patents.

Present-day European systems: Slovenia requires evidence of patentability only by the end of a full-term patent's ninth year. Luxembourg and Switzerland offer optional searches. Ireland offers fully examined full-term patents and short-term patents that may be granted without search or substantive examination. Belgium and the Netherlands publish a search and opinion without making grant conditional on a favourable result. European national routes coexist with the regional EPO route.

Other present-day systems: The South African entry refers to its national route, which remains a depository system. The Brazilian, Chinese, Indian, and Russian entries refer to invention patents, the Japanese entry to ordinary patents, and the US entry to non-provisional utility patents. Singapore permits a local search or reliance on an official foreign search, followed by examination in Singapore.

Sources: Historical UK classifications are based on the Patents, Designs, and Trade Marks Act 1883, Patents Act 1902, and Patents Act 1949. Current UK practice is described by the UK Intellectual Property Office. Contemporary classifications use official guidance from the Malta Commerce Department, Slovenian Intellectual Property Office, Luxembourg government, Swiss Federal Institute of Intellectual Property, Intellectual Property Office of Ireland, Belgian FPS Economy, Netherlands Enterprise Agency, French National Institute of Industrial Property, German Patent and Trade Mark Office, Italian Patent and Trademark Office, the European Patent Office, the South African Companies and Intellectual Property Commission, the Brazilian National Institute of Industrial Property, the China National Intellectual Property Administration, the Indian Patent Office, the Japan Patent Office, the Russian Federal Service for Intellectual Property, the Intellectual Property Office of Singapore, and the United States Patent and Trademark Office.

Table 2. Descriptive statistics of patenting and non-patenting firms, classified by patent grant date

Panel A. Patenting Firms

Variable	Pre-Reform (1899–1904)					Post-Reform (1905–1913)				
	N	Mean	Std. dev.	Min	Max	N	Mean	Std. dev.	Min	Max
Total Assets	150	1,901,681	2,498,363	122,959	9,152,975	422	2,446,829	3,885,251	111,698	22,600,000
Net Profit	150	94,853	136,633	-12,524	723,594	422	137,601	328,212	-152,132	3,028,278
Retained Earnings	150	18,128	31,778	-49,762	191,074	422	38,605	84,728	-169,220	840,462
Total Reserves	150	122,155	187,709	0.0	958,273	421	231,854	656,552	-	7,100,000
Issued Share Capital	150	1,126,782	1,497,560	80,000	5,718,670	421	1,391,146	2,415,023	70,000	15,500,000
Total Debt	150	548,934	845,140	1,262	3,383,415	422	691,042	1,079,207	-	8,658,410
Other Debt	150	149,464	213,398	-	1,219,842	422	174,635	243,986	-	2,011,883
Bond Debt	150	399,470	726,173	-	3,200,000	422	516,408	903,929	-	8,047,012
Cash Total	150	69,580	134,359	-	815,662	422	92,649	147,533	-	1,184,712
Total Debt/Total Assets	150	0.25	0.13	0.00	0.62	422	0.28	0.15	0.00	0.93
Bond Debt/Total Assets	150	0.16	0.12	0.00	0.49	422	0.18	0.14	0.00	0.87
Other Debt/Total Assets	150	0.09	0.07	0.00	0.31	422	0.10	0.08	0.00	0.55
Bank Debt/Total Assets	150	0.01	0.02	0.00	0.13	422	0.01	0.02	0.00	0.19
Trade Credit/Total Assets	150	0.08	0.06	0.00	0.31	422	0.09	0.07	0.00	0.53
Maturity Ratio	145	0.03	0.06	-0.22	0.23	414	0.07	0.11	-0.32	0.94
ROE	150	0.10	0.07	-0.03	0.46	421	0.09	0.08	-0.13	0.42
ROA	150	0.06	0.04	-0.01	0.25	422	0.05	0.04	-0.06	0.24
Tobin's Q	150	0.92	0.46	0.31	4.70	419	1.01	3.90	0.00	75.38
Age	150	12.95	9.65	2.00	40.00	422	16.73	11.33	1.00	59.00
London HQ	150	0.49	0.50	0.00	1.00	422	0.40	0.49	0.00	1.00
Family Controlled	150	0.41	0.49	0.00	1.00	422	0.44	0.50	0.00	1.00

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Table 2 (continued). Descriptive statistics of patenting and non-patenting firms, classified by patent grant date

Panel B. Non-Patenting Firms

Variable	Pre-Reform (1899–1904)					Post-Reform (1905–1913)				
	N	Mean	Std. dev.	Min	Max	N	Mean	Std. dev.	Min	Max
Total Assets	767	887,108	1,231,543	54,743	14,700,000	1,170	969,268	1,344,650	76,243	19,900,000
Net Profit	761	50,134	115,069	-57,150	2,339,787	1,161	52,504	140,691	-73,759	2,795,925
Retained Earnings	759	8,698	26,432	-109,323	272,104	1,159	14,242	45,932	-187,729	836,234
Total Reserves	756	54,945	119,977	-	1,650,000	1,147	90,239	258,118	-	5,400,000
Issued Share Capital	767	537,055	701,737	39,930	5,852,240	1,165	565,605	778,703	50,000	10,000,000
Total Debt	767	223,223	391,786	-	3,668,431	1,170	243,634	356,687	-	3,034,491
Other Debt	767	62,796	90,568	-	945,673	1,170	81,819	110,389	-	942,805
Debentures	767	160,428	344,150	-	3,200,000	1,170	161,815	293,436	-	2,250,000
Cash Total	767	33,590	63,713	-	1,008,680	1,170	35,139	78,980	-	1,409,293
Total Debt/Total Assets	767	0.24	0.16	0.00	0.66	1,170	0.24	0.16	0.00	0.78
Bond Debt/Total Assets	767	0.15	0.15	0.00	0.60	1,170	0.15	0.15	0.00	0.61
Other Debt/Total Assets	767	0.08	0.08	0.00	0.55	1,170	0.09	0.09	0.00	0.49
Bank Debt/Total Assets	767	0.01	0.02	0.00	0.27	1,170	0.01	0.04	0.00	0.37
Trade Credit/Total Assets	767	0.07	0.08	0.00	0.55	1,170	0.08	0.08	0.00	0.49
Maturity Ratio	732	0.04	0.09	-0.42	0.92	1,121	0.06	0.18	-0.57	2.28
ROE	761	0.09	0.10	-0.36	0.96	1,156	0.09	0.10	-0.12	1.50
ROA	761	0.06	0.05	-0.29	0.49	1,161	0.05	0.05	-0.07	0.43
Tobin's Q	767	1.26	5.26	0.17	63.72	1,166	0.99	3.25	0.00	55.98
Age	767	13.88	12.06	0.00	68.00	1,170	19.95	13.03	0.00	76.00
London HQ	767	0.59	0.49	0.00	1.00	1,170	0.59	0.49	0.00	1.00
Family Controlled	767	0.36	0.48	0.00	1.00	1,170	0.38	0.49	0.00	1.00

Notes: This table shows the annual summary statistics of patenting and non-patenting firms before the introduction of patent examination (1899–1904) and after the introduction of patent examination in Britain in 1905 (1905–1913). 'Pre-reform', firms are only categorised as a 'patenting' firm once they have been granted an *unexamined* patent. 'Post-reform', firms are only categorised as a 'patenting' firm once they have been granted an *examined* patent. Mean, standard deviation, minimum, maximum and total number of firm-year observations are shown. See Appendix Table **??** for variable definitions and sources.

Table 3. Descriptive statistics of full sample, 1899–1913

Variable	N	Mean	Std. dev.	Min	Max
Total Assets	2,509	1,248,413	2,134,057	54,743	22,600,000
Net Profit	2,494	68,727	183,374	-152,132	3,028,278
Retained Earnings	2,490	16,915	50,723	-187,729	840,462
Total Reserves	2,474	105,488	337,971	0	7,100,000
Issued Share Capital	2,503	729,341	1,285,480	39,930	15,500,000
Total Debt	2,509	330,898	613,098	0.00	8,658,410
Other Debt	2,509	95,659	150,085	0.00	2,011,883
Debentures	2,509	235,239	513,748	0.00	8,047,012
Cash Total	2,509	46,397	96,824	0.00	1,409,293
Total Debt/Total Assets	2,509	0.25	0.16	0.00	0.93
Bond Debt/Total Assets	2,509	0.16	0.14	0.00	0.87
Other Debt/Total Assets	2,509	0.09	0.08	0.00	0.55
Bank Debt/Total Assets	2,509	0.01	0.03	0.00	0.37
Trade Credit/Total Assets	2,509	0.08	0.07	0.00	0.55
Maturity Ratio	2,412	0.05	0.14	-0.57	2.28
ROE	2,488	0.09	0.09	-0.36	1.50
ROA	2,494	0.05	0.05	-0.29	0.49
Tobin's Q	2,502	1.07	3.99	0.00	75.38
Age	2,509	17.14	12.59	0	76
London HQ	2,509	0.55	0.50	0	1
Family Controlled	2,509	0.39	0.49	0	1

Notes: This table shows the annual summary statistics of all firms in our dataset from 1899–1913. Mean, standard deviation, minimum, maximum and total number of firm-year observations are shown. See Appendix Table **??** for variable definitions and sources.

Table 4. Comparison of average debt ratios, pre- versus post-reform

Panel A. Patent Grant Date

Variable	Firm group	Pre-reform	Post-reform	Difference	(N)
Total Debt/Total Assets	Patenting Firms	0.2483	0.2767	0.0284**	572
	Non-Patenting Firms	0.2374	0.2441	0.0066	1,937
Bond Debt/Total Assets	Patenting Firms	0.1609	0.1784	0.0175*	572
	Non-Patenting Firms	0.1540	0.1512	-0.0028	1,937
Other Debt/Total Assets	Patenting Firms	0.0874	0.0984	0.0109*	572
	Non-Patenting Firms	0.0834	0.0929	0.0095*	1,937
Bank Debt/Total Assets	Patenting Firms	0.0081	0.0068	-0.0012	572
	Non-Patenting Firms	0.0086	0.0130	0.0044***	1,937
Trade Credit/Total Assets	Patenting Firms	0.0788	0.0914	0.0125**	572
	Non-Patenting Firms	0.0731	0.0783	0.0043	1,937

Panel B. Patent Filing Date

Variable	Firm group	Pre-reform	Post-reform	Difference	(N)
Total Debt/Total Assets	Patenting Firms	0.2399	0.2765	0.0365***	655
	Non-Patenting Firms	0.2390	0.2431	0.0041	1,854
Bond Debt/Total Assets	Patenting Firms	0.1534	0.1763	0.0229**	655
	Non-Patenting Firms	0.1556	0.1511	-0.0045	1,854
Other Debt/Total Assets	Patenting Firms	0.0865	0.1002	0.0136**	655
	Non-Patenting Firms	0.0835	0.0920	0.0085**	1,854
Bank Debt/Total Assets	Patenting Firms	0.0074	0.0082	0.0008	655
	Non-Patenting Firms	0.0088	0.0126	0.0038***	1,854
Trade Credit/Total Assets	Patenting Firms	0.0787	0.0918	0.0131**	655
	Non-Patenting Firms	0.0737	0.0777	0.0040	1,854

Notes: This table compares average debt-to-assets ratios for patenting and non-patenting firms before and after the introduction of patent examination in 1905. 'Pre-reform' covers 1899–1904 and 'Post-reform' covers 1905–1913. 'Bond Debt' comprises corporate bonds and debentures; 'Other Debt' is debt excluding bond borrowing; 'Bank Debt' is bank borrowing; and 'Trade Credit' is credit received from trade. Panel A identifies patenting firms using patent grant dates; Panel B uses patent application dates. See Appendix Table **??** for variable definitions and sources. Stars indicate the significance of difference-of-means tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

Table 5. Effect of patent examination on access to total debt

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patenting Firm	-0.008	-0.008	-0.008	0.006	-0.009	-0.009	-0.010	-0.010
	(0.009)	(0.008)	(0.007)	(0.014)	(0.008)	(0.007)	(0.008)	(0.008)
Examination Years	0.001	0.035	0.035	0.035	0.013	0.013	0.030	0.021
	(0.012)	(0.099)	(0.091)	(0.099)	(0.101)	(0.092)	(0.100)	(0.100)
Patenting x Examination	0.022*	0.024**	0.024***	0.024*	0.024**	0.024***	0.024**	0.027**
	(0.012)	(0.011)	(0.007)	(0.013)	(0.011)	(0.007)	(0.011)	(0.011)
Anticipation Patenting				-0.015				
				(0.015)				
Anticipation Years				0.006				
				(0.038)				
Anticipation Effect				-0.004				
				(0.013)				
ROA		-0.393***	-0.393***	-0.393***	-0.433***	-0.433***	-0.429***	-0.318***
		(0.070)	(0.047)	(0.070)	(0.069)	(0.050)	(0.069)	(0.065)
Total Reserves/Total Assets		-0.466***	-0.466***	-0.464***	-0.459***	-0.459***	-0.461***	
		(0.070)	(0.034)	(0.070)	(0.072)	(0.035)	(0.072)	
(Total Reserves + Retained Earnings)/Total Assets								-0.390***
								(0.066)
Age		-0.000	-0.000	-0.000	0.001	0.001	-0.000	0.001
		(0.007)	(0.007)	(0.007)	(0.008)	(0.007)	(0.008)	(0.008)
ln (Issued Share Capital)		-0.084***	-0.084***	-0.084***	-0.091***	-0.091***	-0.095***	-0.086***
		(0.021)	(0.009)	(0.021)	(0.021)	(0.009)	(0.022)	(0.022)
Maturity Ratio					0.036	0.036	0.032	
					(0.039)	(0.026)	(0.039)	
Tobin's Q							0.000	
							(0.000)	
Constant	0.254***	1.381***	1.381***	1.379***	1.452***	1.452***	1.521***	1.384***
	(0.008)	(0.275)	(0.135)	(0.278)	(0.278)	(0.138)	(0.285)	(0.288)
Observations	2,509	2,469	2,469	2,469	2,391	2,391	2,386	2,467
R-squared	0.028	0.143	0.143	0.144	0.150	0.150	0.153	0.129
Number of Firms	295	293	293	293	287	287	287	292
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Clustered SE	YES	YES	NO	YES	YES	NO	YES	YES

Notes: This table presents results of two-way fixed effects difference-in-differences panel regressions examining the introduction of patent examination on patenting firms' debt financing. The dependent variable is Total Debt/Total Assets. 'Patenting Firm' is a dummy variable that =1 in years once a firm has obtained a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. 'Anticipation Effect' is an interaction to test any additional anticipatory effect of patenting after the announcement of patent examination legislation, but before its implementation in 1905 (i.e., covering 1903 and 1904). See Appendix Table **??** for variable definitions and sources. (Clustered) robust standard errors (at the firm level) in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table 6. Effect of patent examination on access to capital market (bond) debt

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Patenting Firm	-0.017*	-0.017**	-0.017***	-0.017*	-0.017***	-0.018**
	(0.009)	(0.008)	(0.006)	(0.009)	(0.006)	(0.009)
Examination Years	-0.009	0.008	0.008	0.002	0.002	-0.001
	(0.010)	(0.074)	(0.071)	(0.077)	(0.073)	(0.076)
Patenting x Examination	0.029**	0.031***	0.031***	0.030**	0.030***	0.033***
	(0.013)	(0.012)	(0.005)	(0.012)	(0.005)	(0.012)
ROA		-0.274***	-0.274***	-0.287***	-0.287***	-0.228***
		(0.056)	(0.037)	(0.060)	(0.039)	(0.052)
Total Reserves/Total Assets		-0.285***	-0.285***	-0.270***	-0.270***	
		(0.059)	(0.027)	(0.059)	(0.027)	
(Total Reserves + Retained Earnings)/Total Assets						-0.237***
						(0.055)
Age		0.000	0.000	0.001	0.001	0.001
		(0.006)	(0.005)	(0.006)	(0.005)	(0.006)
ln (Issued Share Capital)		-0.065***	-0.065***	-0.070***	-0.070***	-0.066***
		(0.022)	(0.007)	(0.023)	(0.007)	(0.022)
Maturity Ratio				0.020	0.020	
				(0.038)	(0.021)	
Constant	0.161***	1.020***	1.020***	1.084***	1.084***	1.021***
	(0.008)	(0.281)	(0.106)	(0.289)	(0.108)	(0.288)
Observations	2,509	2,469	2,469	2,391	2,391	2,467
R-squared	0.022	0.109	0.109	0.110	0.110	0.099
Number of Firms	295	293	293	287	287	292
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Clustered SE	YES	YES	NO	YES	NO	YES

Notes: This table presents results of two-way fixed effects difference-in-differences panel regressions examining the introduction of patent examination on patenting firms' debt financing. The dependent variable is Bond Debt/Total Assets. 'Patenting Firm' is a dummy variable that =1 in years once a firm has obtained a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. See Appendix Table **??** for variable definitions and sources. (Clustered) robust standard errors (at the firm level) in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table 7. Effect of patent examination on access to non-capital market (other) debt, aggregated

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Patenting Firm	0.009	0.009	0.009*	0.008	0.008	0.008
	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)	(0.005)
Examination Years	0.010	0.027	0.027	0.011	0.011	0.022
	(0.008)	(0.072)	(0.070)	(0.072)	(0.072)	(0.071)
Patenting x Examination	-0.007	-0.007	-0.007	-0.006	-0.006	-0.006
	(0.006)	(0.006)	(0.005)	(0.006)	(0.005)	(0.006)
ROA		-0.119**	-0.119***	-0.146***	-0.146***	-0.090*
		(0.051)	(0.036)	(0.054)	(0.039)	(0.048)
Total Reserves/Total Assets		-0.181***	-0.181***	-0.189***	-0.189***	
		(0.051)	(0.026)	(0.053)	(0.027)	
(Total Reserves + Retained Earnings)/Total Assets						-0.153***
						(0.046)
Age		-0.000	-0.000	0.001	0.001	-0.000
		(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
ln (Issued Share Capital)		-0.019	-0.019***	-0.021	-0.021***	-0.020
		(0.012)	(0.007)	(0.013)	(0.007)	(0.012)
Maturity Ratio				0.016	0.016	
				(0.024)	(0.020)	
Constant	0.092***	0.361**	0.361***	0.368**	0.368***	0.363**
	(0.005)	(0.163)	(0.104)	(0.170)	(0.107)	(0.165)
Observations	2,509	2,469	2,469	2,391	2,391	2,467
R-squared	0.023	0.049	0.049	0.053	0.053	0.045
Number of Firms	295	293	293	287	287	292
Time FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Clustered SE	YES	YES	NO	YES	NO	YES

Notes: This table presents results of two-way fixed effects difference-in-differences panel regressions examining the introduction of patent examination on patenting firms' debt financing. The dependent variable is Non-Capital Market Debt/Total Assets. 'Patenting Firm' is a dummy variable that =1 in years once a firm has obtained a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. See Appendix Table **??** for variable definitions and sources. (Clustered) robust standard errors (at the firm level) in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table 8. Effect of patent examination on access to non-capital market (other) debt, disaggregated by debt type

Panel A. Bank Debt

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Patenting Firm	0.003*	0.003*	0.003	0.003*	0.003	0.003*
	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)
Examination Years	0.003	0.039	0.039	0.041	0.041	0.038
	(0.006)	(0.026)	(0.040)	(0.027)	(0.041)	(0.025)
Patenting x Examination	-0.007**	-0.007**	-0.007**	-0.007**	-0.007**	-0.007**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
ROA		-0.069***	-0.069***	-0.067***	-0.067***	-0.060***
		(0.021)	(0.020)	(0.023)	(0.022)	(0.020)
Total Reserves/Total Assets		-0.052**	-0.052***	-0.053**	-0.053***	
		(0.026)	(0.015)	(0.026)	(0.015)	
(Total Reserves + Retained Earnings)/Total Assets						-0.047*
						(0.024)
Age		-0.002	-0.002	-0.003	-0.003	-0.002
		(0.002)	(0.003)	(0.002)	(0.003)	(0.002)
ln (Issued Share Capital)		-0.009	-0.009**	-0.008	-0.008*	-0.009
		(0.007)	(0.004)	(0.007)	(0.004)	(0.007)
Maturity Ratio				-0.006	-0.006	
				(0.012)	(0.012)	
Constant	0.010***	0.149*	0.149**	0.143	0.143**	0.151*
	(0.004)	(0.084)	(0.059)	(0.088)	(0.061)	(0.084)
Observations	2,509	2,469	2,469	2,391	2,391	2,467
R-squared	0.015	0.026	0.026	0.026	0.026	0.026
Number of Firms	295	293	293	287	287	292
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Clustered SE	YES	YES	NO	YES	NO	YES

Continued on next page...

Table 8 (continued). Effect of patent examination on access to non-capital market (other) debt, disaggregated by debt type

Panel B. Trade Credit

Variable	(7)	(8)	(9)	(10)	(11)	(12)
Patenting Firm	0.004	0.004	0.004	0.004	0.004	0.002
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)
Examination Years	0.007	-0.022	-0.022	-0.041	-0.041	-0.011
	(0.007)	(0.064)	(0.061)	(0.065)	(0.063)	(0.009)
Patenting x Examination	0.001	0.002	0.002	0.003	0.003	0.004
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
ROA		-0.037	-0.037	-0.062	-0.062*	-0.016
		(0.045)	(0.032)	(0.047)	(0.034)	(0.042)
Total Reserves/Total Assets		-0.135***	-0.135***	-0.143***	-0.143***	
		(0.037)	(0.023)	(0.039)	(0.023)	
(Total Reserves + Retained Earnings)/Total Assets						-0.112***
						(0.034)
Age		0.003	0.003	0.004	0.004	-0.011
		(0.005)	(0.005)	(0.005)	(0.005)	(0.009)
ln (Issued Share Capital)		-0.010	-0.010	-0.012	-0.012*	-0.026
		(0.009)	(0.006)	(0.009)	(0.006)	(0.064)
Maturity Ratio				0.017	0.017	0.004
				(0.020)	(0.018)	(0.005)
Constant	0.081***	0.194	0.194**	0.207*	0.207**	0.195
	(0.004)	(0.122)	(0.091)	(0.125)	(0.093)	(0.123)
Observations	2,509	2,469	2,469	2,391	2,391	2,467
R-squared	0.015	0.031	0.031	0.036	0.036	0.028
Number of Firms	295	293	293	287	287	292
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Clustered SE	YES	YES	NO	YES	NO	YES

Notes: Results are from two-way fixed-effects difference-in-differences panel regressions. The dependent variable is Bank Debt/Total Assets in Panel A and Trade Credit/Total Assets in Panel B. 'Patenting Firm' equals one once a firm has obtained an *unexamined* patent before 1905 or an *examined* patent from 1905 onwards. 'Examination Years' equals one from 1905 onwards. See Appendix Table **??** for variable definitions and sources. (Clustered) robust standard errors at the firm level are in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table 9. Effect of patent examination on the cost of debt, and on equity financing

Variable	(1) Weighted Average Debt Cost	(2) Total Common/ Total Assets	(3) Total Preference/ Total Assets
Patenting Firm	0.033	0.002	0.011*
	(0.053)	(0.006)	(0.006)
Examination Years	-0.534	-0.009	0.059
	(0.814)	(0.053)	(0.054)
Patenting x Examination	0.042	-0.013*	-0.018**
	(0.030)	(0.007)	(0.008)
ROA	-0.494*	-0.170***	-0.063
	(0.286)	(0.064)	(0.044)
Total Reserves/Total Assets	-0.044	-0.174***	-0.093*
	(0.457)	(0.066)	(0.052)
Age	0.038	-0.002	-0.005
	(0.056)	(0.004)	(0.004)
ln (Issued Share Capital)	-0.134	0.063***	0.048*
	(0.195)	(0.023)	(0.025)
Constant	5.646**	-0.378	-0.336
	(2.649)	(0.298)	(0.316)
Observations	1,595	2,469	2,469
R-squared	0.010	0.156	0.068
Number of Firms	206	293	293
Year FE	YES	YES	YES
Firm FE	YES	YES	YES
Clustered SE	YES	YES	YES

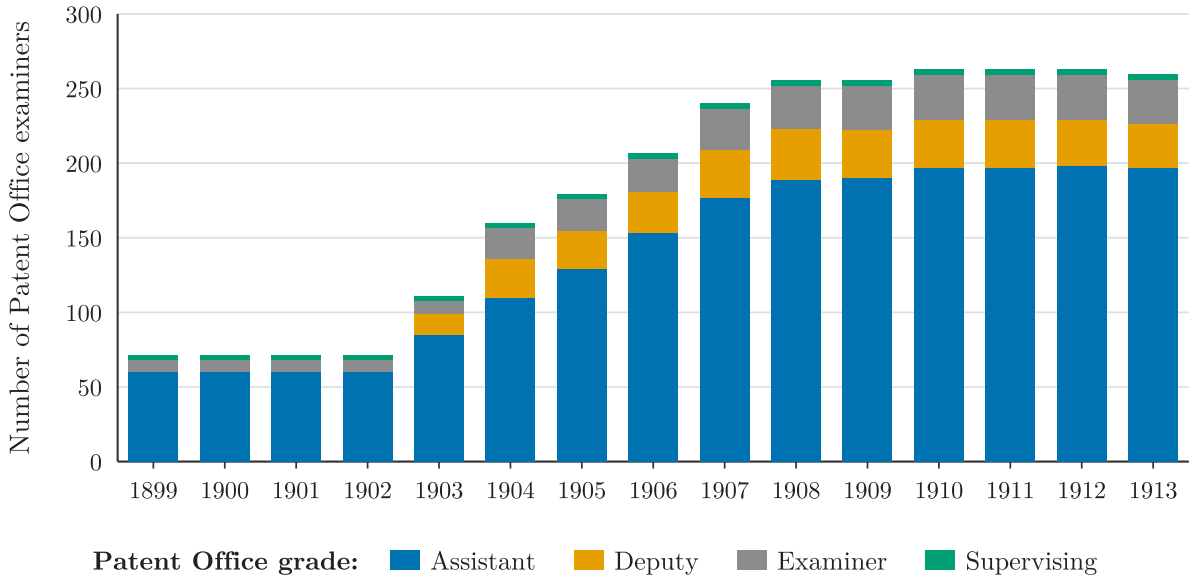
Notes: This table presents results of two-way fixed-effects difference-in-differences panel regressions examining the introduction of patent examination on patenting firms' financing. The dependent variable is the weighted average coupon payment on a firm's issued bonds in column (1); Total Common Share Capital/Total Assets in column (2); and Total Preference Share Capital/Total Assets in column (3). 'Patenting Firm' is a dummy variable that =1 in years once a firm has obtained a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. See Appendix Table **??** for variable definitions and sources. Clustered robust standard errors at the firm level are in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table 10. Effect of patent examination on firm size

Variable	(1)	(2)	(3)	(4)	(5)
Patenting Firm	-0.041*	-0.042*	-0.037	-0.046*	-0.029*
	(0.022)	(0.023)	(0.025)	(0.026)	(0.015)
Examination Years	0.226***	-0.193	-0.223	-0.185	-0.121
	(0.037)	(0.149)	(0.154)	(0.158)	(0.105)
Patenting x Examination	0.098***	0.103***	0.106***	0.113***	0.067***
	(0.031)	(0.031)	(0.032)	(0.033)	(0.018)
ROA		0.005	0.145	-0.042	0.233*
		(0.212)	(0.166)	(0.204)	(0.120)
Total Reserves/Total Assets		0.202			
		(0.185)			
ln (Total Reserves)			0.038***	0.033***	0.029***
			(0.011)	(0.010)	(0.007)
ln (Retained Earnings)				0.021***	0.013***
				(0.007)	(0.004)
ln (Issued Share Capital)					0.710***
					(0.028)
Age		0.028**	0.029**	0.025**	0.014*
		(0.012)	(0.012)	(0.013)	(0.008)
Constant	13.281***	12.993***	12.648***	12.579***	3.692***
	(0.025)	(0.126)	(0.142)	(0.147)	(0.364)
Observations	2,509	2,471	2,165	2,036	2,035
R-squared	0.173	0.168	0.254	0.272	0.664
Number of Firms	295	293	280	264	264
Year FE	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES
Clustered SE	YES	YES	YES	YES	YES

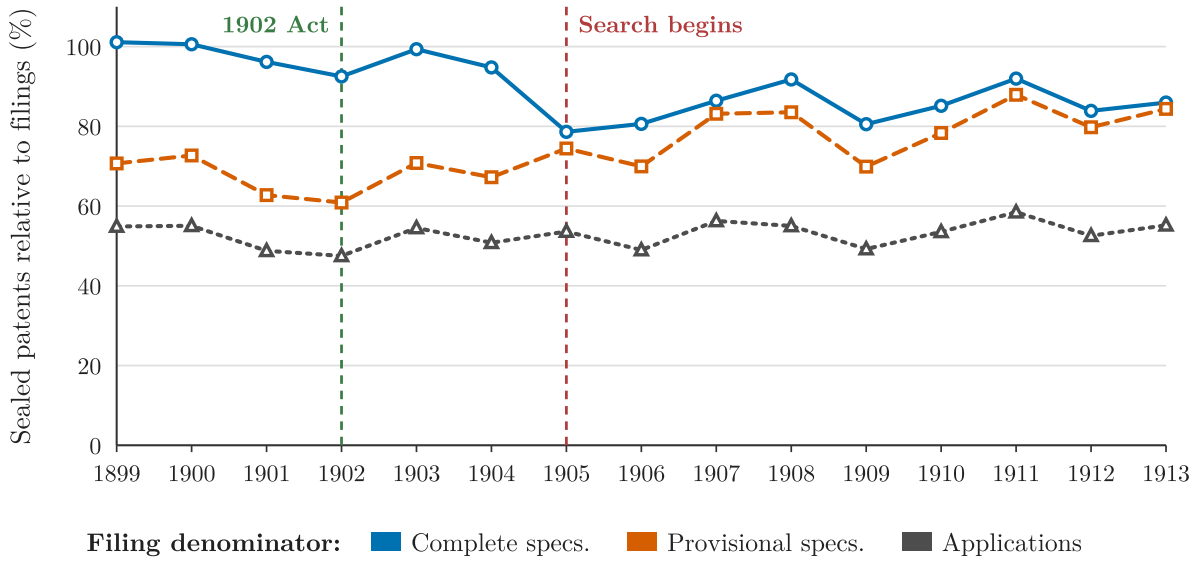
Notes: This table presents results of two-way fixed effects difference-in-differences panel regressions examining the introduction of patent examination on patenting firms' debt financing. The dependent variable is the natural log of the book value of Total Assets. 'Patenting Firm' is a dummy variable that =1 in years once a firm has obtained a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. See Appendix Table **??** for variable definitions and sources. Clustered robust standard errors (at the firm level) in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Figure 1. Patent examiners employed by the Patent Office, 1899–1913



Sources: Authors' calculations using data published in the *Reports of the Comptroller-General* (1899–1913).

Figure 2. Patenting activity in the UK, 1899–1913



Notes: The series divide patents sealed in each calendar year by complete specifications, provisional specifications, and all applications filed in that year. They are annual ratios rather than cohort-based sealing rates. The green line marks Royal Assent of the Patents Act 1902 (18 December 1902); the red line marks the introduction of prior-art search examination (1 January 1905).

Sources: Authors' calculations using data published in the *Reports of the Comptroller-General* (1899–1913).

SUPPLEMENTARY MATERIALS

APPENDIX A

Contemporary press evidence on the corporate value of patents

This supplementary appendix collates contemporary press evidence from *The Times* (of London) and *The Financial Times* highlighting the value of patents to firms and investors, and their role in securing bondholders, during 1899–1913. It includes company prospectuses, partnership and investment advertisements, AGM reports, and court cases. These materials show that patents already functioned as signals of firm value before systematic examination.

Before 1905, firms seeking finance sometimes commissioned patent agents, counsel, and technical experts to search earlier specifications and assess the novelty or validity of their patents. Favourable opinions could then be reproduced in prospectuses as private certification of firms' claims. This arrangement had important limitations. Firms selected and paid the experts, no common standard governed their assessments, and unfavourable findings could be withheld. Investors therefore had to evaluate both the patent and the expert, while conflicts of interest, manipulation, and corruption could further weaken the signal. Patent examination replaced this patchy, issuer-funded system with a standardised public record, and contemporary press commentary explicitly connected the new process to the saleable value of patents.

A legislative roundup published in *The Times* argued that prior-art search would increase inventors' confidence in being the '"true and first inventor", and enhance the saleable value of the patent', while cautioning that examination was 'nevertheless, not in any way to guarantee the legal validity of the patent' (15 January 1903). The commentary thus distinguished the informational value of official scrutiny from a guarantee of legal validity. Prior-art search could make a patent more credible and saleable without ensuring that it would survive a subsequent court challenge.

Table A1 reports examples from company prospectuses issued when raising capital before and after the 1905 reform. Some concern IPOs by companies incorporated to acquire and commercialise existing patent rights, making those rights and underlying inventions central to their value propositions. Others sought expansion or working capital while highlighting patented technology to investors. Several prospectuses explicitly referred to patent examination or the use of patent rights as security for debenture holders.

[Table A1 about here]

Many prospectuses sought to support claims about the quality of patented inventions using testimony from patent experts, typically a Chartered Patent Agent or eminent Patent Counsel. Such testimony drew on the expert's reputational capital and could make a firm's claims more credible. A common approach was to report that a patent agency had searched earlier British specifications, followed by counsel's opinion on whether the patent would survive a legal challenge. However, the firm selected and paid the experts, determined the scope of the search, and controlled which findings appeared in the prospectus. Private certification could therefore add information to the patent signal, but investors could not readily distinguish credible expert assessment from issuer-funded promotion.

For example, the *Pedlar Brake and Two-Speed Gear Company* reported that it incorporated to acquire British and foreign patent rights for bicycle technology. To demonstrate the strength of its patents, it hired a patent agency, *Messrs Rayner and Co.,* to conduct an exhaustive search of prior British patent specifications, and Mr J. Fletcher Moulton an MP and recognised authority on patent law, for legal opinion. The findings were reported in the prospectus:

> 'Dear Sir,- In accordance with your instructions we have made a search in the Records of the British Patent Office on the subject and claims of the complete specification of the above application, the search extending from the date of the application back to and including the year 1877 under the headings of Velocipedes: driving mechanism; Motor Road Vehicles: driving, gear, foot, hand or power. During the course of the search nearly 4,000 specifications were examined, and as the result we are pleased to be able to inform you that we consider the apparatus of Mr. Sturgess to possess ample novelty to support a good and valid Patent. None of the previous patents show such a combination of advantages as Mr. Sturgess's hub contains and give so complete a control over the wheel.'
>
> *Rayner & Co.* (28 September 1904)

> 'I am of opinion that with the suggested amendments, which have been evidently drawn with great care, the Patents submitted to me, namely, Nos. 1127 of 1900, 24965 of 1902, and 21640 of 1903, will be good and valid, and will not be anticipated by any of the prior specifications sent with the case.'
>
> *J. Fletcher Moulton, Temple, E.C.* (21 December 1904)

In the example of the 1900 prospectus of *The Lanston Monotype Corporation*, the firm explicitly describes its patent right as a form of collateral to prospective bondholders:

> 'The Debenture Stock will be secured to the Trustees for the Debenture Stock Holders by a First Mortgage on the freehold and leasehold properties and the Letters Patent, and a floating charge on the undertaking and general assets of the Corporation.'

The Lanston Monotype Corporation (21 December 1900)

Notably, in a later (post 1905) prospectus, the *British Coalite Company* explicitly referenced that the validity of their patent was confirmed by the new British patent examination system:

> 'Exhaustive searches as to previous patents have been made in the British, German, French, and United States Patent Records, by Messrs Edward Evans and Co., of Chancery Lane, W.C., which have been confirmed by the official examinations as to novelty made in the United Kingdom, Germany, and the United States.'
>
> *The British Coalite Company* (18 June 1907)

Patent rights were also used to attract investors in the form of small-scale startup capital or business partners. Notices frequently appeared in *The Times* newspaper's "*Partnerships, Investments, &c.*" column. For example, an advertisement dated 12 November 1907 states: 'ENGINEERING FIRM, protected by several patents, REQUIRES further WORKING CAPITAL, privately or syndicate. Suit naval or military office, or millowner'. Another dated 11 July 1907, requests: 'FINANCIER, with £1,000 WANTED, to place a very successful patent on a steam plant. One who would not object to a company being formed'. Notably, these advertisements disclose little about the patents themselves beyond asserting ownership. Yet, this minimal disclosure appears to have been deemed sufficient to attract interested investors, suggesting that patent ownership alone was believed to serve as a credible signal of value to potential investors.

Patent ownership was not, however, accepted uncritically. The financial press sometimes acted as an informal monitor by drawing investors' attention to evidently weak claims. In 1899, *The Financial Times* questioned how a method of insurance investment could be patentable, describing the company involved as a 'pig-in-a-poke' and its concept as 'bunkum not deserving the least consideration from investors' (22 August 1899, p. 2). Such scrutiny could warn investors but could not determine a patent's validity. Without official prior-art examination, weak patents could retain apparent validity unless challenged through costly litigation, leaving firms and investors to bear the expense of establishing their legal validity.

At the 1899 AGM of the British Motor Company, prior to the introduction of patent examination, both the strategic value of patents and the difficulty of proving their validity were emphasised. It was reported that patents would allow the company to "cut the ground from beneath the feet of its rivals", and that legal challenges, necessary to prove the validity of their patents, had "added immensely to their value", as every victory "sent the price up". The

Table A1. Examples of prospectuses published in *The Times* mentioning patents

Company	Year	Capital Issued	Reason for Raising Capital
Nernst Electric Light	1899	Shares	Acquisition of patents
The British Charrier Wood Carving Company*	1899	Shares	Acquisition of patents
Lanston Monotype Corporation*	1900	Debentures	Expansion of capital stock
Bayliss, Jones, and Bayliss*	1901	Debentures	Acquisition of land and patents
The Linotype Company	1901	Shares and Debentures	Expansion of capital stock
The Projectile Company (1902)*	1902	Shares and Debentures	Working capital
The Peldar Brake and Two-Speed Gear Company	1905	Shares	Working capital
The British Coalite Company	1907	Shares	Working capital
The Molassine Company*	1907	Shares	Working capital
Vickers, Sons & Maxim, Limited*	1908	Shares and Debentures	Working capital
Aron Electricity Meter*	1911	Debentures	Expansion of capital stock

Notes: * indicates companies in our dataset.

company saw legal enforcement to patent infringement as a necessary defence against those who would "steal the shareholder's property" (*The Financial Times*, 3 January 1899, p. 5).

The role of patents as valuable firm assets is also evident in legal proceedings following the liquidation of failed companies, particularly in their use as security for bondholders. In the case of Celladema (Limited), the original purchase price of the company reportedly considered that their patent was worth £5,000 (*The Times*, 23 February 1899, p. 14). Following the winding-up order of the London and Brimingham Brass Foundry Ltd, initiated by debenture holders, it was noted that the company was formed specifically to acquire another firm's patent rights, for which £40,000 had been paid (*The Financial Times*, 3 January 1899, p. 3). In another example, the assets of Fuller, Horsey, Sons and Cassell were listed for auction by order of the Court, instigated by debenture holders. The assets included the company's letters patent and patented bag machines (*The Times*, 17 June 1905, p. 19).

APPENDIX B

Data construction, patent measures, and variable definitions

B.1 Sample construction and matching

We identify patenting firms using the European Patent Office's PATSTAT Biblio database (Autumn 2016 version), which combines patent records from national and regional offices. Although its historical coverage is incomplete, British patent data are comprehensive from the late 1890s onwards (Billington and Hanna, 2021). PATSTAT records patentee names, filing and grant years, patent titles, abstracts, complete specifications, and forward citations. We begin the sample in 1899 to ensure comprehensive coverage and a substantial pre-reform window. We end in 1913 because the outbreak of World War I disrupted firms and financial markets and altered Patent Office reporting.

We manually match patentees to companies listed on the London Stock Exchange using the *Investor's Monthly Manual* (IMM) and *Stock Exchange Daily Official Lists* (SEDOL). IMM is available digitally through the International Center for Finance at the Yale School of Management. SEDOL and the corporate balance sheets used below form part of the Stock Exchange Collection at the London Archives, Guildhall Library. The matching procedure yielded 150 patenting companies classified in SEDOL as either 'Commercial, Industrial' or 'Iron, Coal, Steel'. We restrict the analysis to these categories because they contain the overwhelming majority of officially listed British companies patenting during 1899–1913. The few patenting firms in other categories, including breweries and railways, provide too little within-category variation for meaningful comparisons and operated in sectors where patenting was relatively peripheral.

The initial control group contains 200 non-patenting firms randomly selected from the same two SEDOL categories. We draw from each category in proportion to its representation among patenting companies. This stratified random sampling improves sectoral comparability and reduces the possibility that differences in financing reflect the composition of the treatment and control groups rather than patenting behaviour.

We hand-collect annual balance sheets for every firm for which records survive. Balance sheets are unavailable for nine companies that patented during the sample period: four patented only before 1905, four only after 1905, and one in both periods. The final dataset contains 132 patenting companies and 163 non-patenting companies, yielding 2,509 firm-year observations. Of the patenting firms, 27 patented only before 1905, 71 only from 1905 onwards, and 34 in both periods. Accordingly, 61 firms patented before 1905 and 105 from 1905 onwards.

B.2 Patent measures

The Patent Office's *Reports of the Comptroller-General* provide annual population statistics on the results of examination from 1905 to 1913. Table B1 reports whether complete specifications were wholly, partly, or not anticipated by earlier British specifications and how applicants responded to the search. 'Wholly anticipated' means that all claims matched an earlier specification; 'partly anticipated' means that at least one claim did so. Applicants could amend their specifications upon receiving the search report or after an adverse decision. If they proceeded without adequate amendment, the Patent Office could require the insertion of references to anticipating specifications. The Patent Office ceased publishing these outcomes in comparable detail after the outbreak of World War I.

Table B2 divides the PATSTAT data into a pre-reform period, 1899–1904, and a post-reform period, 1905–1913. Within each period, we distinguish companies and individuals using PATSTAT's applicant classification and separately identify the listed firms in our sample. 'Patent stock' is the number of patents granted to each patentee within the relevant period. The listed firms are frequent users of the patent system: they obtained an average of 16 patents before the reform and 24 afterwards. Excluding British Thomson-Houston, which obtained more than 1,600 patents by 1913, the corresponding averages are 6 and 14.

The 'Reference' variable identifies references contained in the patent specification. Before 1905, these were voluntarily supplied by applicants and varied in coverage. Examination introduced a systematic official search and allowed the Patent Office to add a forced reference if the applicant did not adequately address anticipating specifications. Listed firms supplied references more often than other applicants before 1905, but none of their post-reform patents contained a forced reference. This may indicate that sophisticated applicants generally amended or withdrew weak claims before sealing. The rise in applicant-supplied references among individuals is also consistent with evidence that examiner-added references can induce applicants to disclose more prior art themselves (Allison and Hunter, 2006).

'Not granted' identifies applications for which a complete specification was filed but the sealing fee was not paid. Before 1905, there were few discernible instances of applicants abandoning applications. After 1905, abandonment rose to around one per cent among companies and two per cent among individuals. Because the additional £1 sealing fee was payable only after applicants received the search report, this measure captures withdrawal after examination rather than fee-based selection at entry.

Table B3 reports annual patent applications and forward citations for the listed patenting firms, excluding British Thomson-Houston. These firms collectively filed 1,859 applications.

Both the number of firms filing patents and the total volume of applications were higher after 1905. Applications filed following the introduction of examination also received more forward citations on average, with the increase visible across several years, particularly from 1909 onwards. These statistics are descriptive, but they are consistent with an improvement in the average quality and informational value of the patent pool.

B.3 Financial data and variable construction

We transcribe balance-sheet items from firms' annual accounts and combine them with company characteristics from the *Stock Exchange Yearbook*, SEDOL, and Grace's Guide. Table B4 defines the treatment variables, balance-sheet items, firm characteristics, financial ratios, and variables used in robustness exercises. The main debt categories follow the terminology in firms' accounts. Bond debt comprises debentures and debenture stock; other debt is total debt excluding bond debt; bank debt records borrowing from banks; and trade credit records amounts owed to suppliers.

Historical financial statements require some caution because uniform statutory accounting standards were introduced only later. Questions have been raised about the accuracy of company accounts before the Companies Act 1948, although the principal concerns generally relate to periods after our sample (Aldous et al., 2023, p. 143). The presence of both young and mature companies does not itself indicate a sampling problem: early-stage and startup companies commonly listed on the British stock exchange during this period (Acheson et al., 2016; Fjesme et al., 2019).

The detailed statistics in Table 2 show that, before examination, patenting and non-patenting firms had average total debt-to-assets ratios of 0.25 and 0.24 and maxima of 0.62 and 0.66. After 1905, the corresponding average for patenting firms increased to 0.28 while that for controls remained unchanged; the maxima increased to 0.93 and 0.78. Before examination, patenting and non-patenting firms had average bond debt-to-assets ratios of 0.16 and 0.15, respectively. Both groups had bank debt-to-assets ratios of 0.01, while their trade credit-to-assets ratios were 0.08 and 0.07. After 1905, trade credit increased by 0.01 in both groups and bank borrowing remained constant, whereas the bond debt-to-assets ratio increased from 0.16 to 0.18 among patenting firms and remained unchanged among controls. Table 3 additionally shows that patenting and non-patenting firms had similar average ROA and ROE, although patenting firms were larger and somewhat more likely to be family or director controlled, while controls were somewhat more likely to be headquartered in London.

Table B1. Results of patent examination, 1905–1913

Year	Sealed	Prior art search			Amendments following search		
		Anticipated		Not anticipated	Amended by patentee		Insertion of forced reference
		Wholly	Partly		Without hearing	After decision	
1905	15,369	1,308	8,238	5,283	7,725	1,060	556
1906	16,972	1,111	9,743	6,118	8,440	1,544	553
1907	16,679	938	10,201	5,540	8,674	1,800	424
1908	16,566	865	10,462	5,239	8,931	1,760	310
1909	16,973	1,048	10,552	5,373	10,053	1,029	379
1910	17,082	1,185	10,383	5,514	10,391	747	322
1911	16,724	1,193	10,233	5,298	10,337	679	303
1912	17,407	1,174	10,903	5,830	11,003	650	311
1913	16,708	1,002	10,401	5,305	10,422	593	286

Notes: 'Sealed Patents' refer to all patents where a complete specification has been granted and the sealing fee paid. 'Wholly' anticipated refers to a patent's claims entirely matching a previous specification. 'Partly' anticipated refers to one or several of a patent's claims matching previous specifications.

Sources: Authors' calculations using data published in the *Reports of the Comptroller-General* (1905–1913).

Table B2. Summary statistics of all British patenting activity, 1899–1913

Variable	Pre-Reform						Post-Reform					
	Individuals (57,277)		Firms (3,107)		Sample Firms (61)		Individuals (86,357)		Firms (7,189)		Sample Firms (105)	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Patent Stock	1.80	10.14	2.31	13.02	16.66 *(6.08)*	82.47 *(9.81)*	1.95	7.36	2.91	15.26	24.16 *(14.37)*	101.77 *(19.47)*
Backward Citation	4%	17%	8%	24%	15%	36%	9%	29%	9%	29%	19%	39%
Void	-	-	-	-	-	-	0%	1%	0%	1%	0%	0%
Not Granted	0%	1%	0%	3%	0%	0%	3%	16%	1%	11%	0%	4%
Forward Citation	0.14	0.50	0.14	0.45	0.06	0.31	0.25	0.90	0.25	0.79	0.19	0.61
Forced Reference	-	-	-	-	-	-	2%	15%	1%	10%	0%	4%

Notes: 'Pre-reform' covers 1899–1904; 'post-reform' covers 1905–1913. 'Individuals' refers to patents held by a person, 'Firms' refers to patents held by any kind of commercial firm, and 'Sample Firms' refers to our sample of LSE-listed patenting firms. 'Patent stock' is the total number of patents held by each unique person or firm in the period in question. The numbers in parentheses represent the patent stock when the 'British Thomson-Houston' company has been removed, as it is an outlier, having obtained over 1,000 patents in our period of observation. 'Backward citation' is a crude measure of how many references each complete specification makes to earlier patents (e.g., '7%' means that 7% of patents granted had at least one backward citation). 'Void' is the percentage of patents indicating that the patent was void in the abstract text. 'Not Granted' refers to the percentage of patents for which the sealing fee was listed as not paid in the abstract text; these patents were abandoned following prior-art search examination. 'Forward Citation' is the average number of forward citations from future patents received by each patent. 'Forced Reference' is the percentage of patents containing backward citations to earlier specifications that the Patent Office required to be appended after the applicant declined to amend the relevant claims.

Sources: Authors' calculations using data from PATSTAT.

Table B3. Patenting statistics of sample firms, 1899–1913

Year	Total Patent Applications					Total Forward Citations				
	Mean	Sum	Max	Min	Count	Mean	Sum	Max	Min	Count
1899	2.72	49	10	1	18	0.67	12	4	0	18
1900	1.61	29	6	1	18	0.06	1	1	0	18
1901	2.30	69	17	1	30	0.10	3	1	0	30
1902	2.46	69	10	1	28	0.14	4	1	0	28
1903	2.59	75	12	1	29	0.38	11	3	0	29
1904	2.55	74	16	1	29	0.34	10	4	0	29
1905	3.26	150	15	1	46	0.59	27	9	0	46
1906	2.52	116	14	1	46	0.52	24	3	0	46
1907	3.08	154	16	1	50	0.50	25	6	0	50
1908	2.59	132	10	1	51	0.45	23	5	0	51
1909	2.93	161	17	1	55	1.04	57	10	0	55
1910	3.73	190	17	1	51	1.20	61	8	0	51
1911	3.46	194	26	1	56	0.86	48	20	0	56
1912	3.48	212	25	1	61	1.07	65	11	0	61
1913	3.78	185	22	1	49	0.98	48	8	0	49
Total	3.01	1,859	–	–	617	0.68	419	–	–	617

Notes: Table reports total patent applications and total forward citations received by those applications for our sample of LSE-listed patenting firms, by filing year. Mean, sum, minimum and maximum are calculated across firms with at least one patent application in the year; count reports the number of firm-year observations. Table excludes patents associated with 'British Thomson-Houston'.

Sources: Authors' calculations using data from PATSTAT.

Table B4. Variable definitions and sources

Variable	Definition	Sources
Patenting Behaviour		
Patenting Firm	=1 in years after a firm has a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards, =0 otherwise (year of accounts)	PATSTAT
Examination Years	=1 if examination regime (1905–1913), =0 otherwise (1899–1904)	PATSTAT
Patenting x Examination	Interaction of Patenting Firm and Examination Years (the 'Examination Effect')	PATSTAT
Anticipation Patenting	=1 in 1903 or 1904 (after examination is announced but before introduction) after a firm has a granted patent, =0 otherwise (year of accounts)	PATSTAT
Anticipation Years	=1 if 1903 or 1904 (i.e., the years after examination announcement, but before policy implementation)	PATSTAT
Anticipation Effect	Interaction of Anticipation Patenting and Anticipation Years	PATSTAT
Balance Sheet Items		
Total Assets	Book value of total assets	Balance sheets
Issued Share Capital	Book value of total issued shares	Balance sheets
Total Debt	Book value of total debt	Balance sheets
Bond Debt	Book value of corporate bonds (debentures)	Balance sheets
Other Debt	Total debt excluding bond debt	Balance sheets
Bank Debt	Total bank borrowing	Balance sheets
Cash Total	Total cash and cash equivalents	Balance sheets
Trade Credit	Total credit from trade	Balance sheets
Total Reserves	Total held in reserves and funds	Balance sheets
Net Profit	Profit before distribution (after expenses including depreciation and interest)	Balance sheets
Retained Earnings	Accumulated net income or loss carried to the following year	Balance sheets
Firm Characteristics		
Age	The number of years since incorporation	SEYB
London HQ	=1 if firm is headquartered in London, =0 otherwise	SEYB
Family Controlled	=1 if family-controlled firm, =0 otherwise (family controlled if at least two listed directors have the same surname, or the managing director's surname is in the company's name)	SEYB
Iron	=1 if firm is in the Iron, Coal & Steel sector, =0 otherwise	SEYB
Industry	Firm's principal industry, assigned using an 18-category classification	Grace's Guide

Continued on next page...

Table B4 (continued). Variable definitions and sources

Variable	Definition	Sources
Balance Sheet Ratios		
Return on Assets (ROA)	Net profit/total assets	Balance sheets
Return on Equity (ROE)	Net profit/total issued share capital (book value)	Balance sheets
Tobin's Q	(Book value of debt + book value of preference equity + market value of common equity)/book value of total assets	Balance sheets; SEDOL
Maturity Ratio	Retained earnings/ordinary equity (book value)	Balance sheets; SEDOL
Robustness Variables		
Weighted Average Debt Cost	Weighted average coupon payments on a firm's issued bonds	Balance sheets
Total Common/Total Assets	Book value of total common shares issued/total assets	Balance sheets
Total Preference/Total Assets	Book value of total preference shares issued/total assets	Balance sheets

Notes: PATSTAT = EPO's PATSTAT Biblio (Autumn 2016 version); SEYB = *Stock Exchange Yearbook*; SEDOL = *Stock Exchange Daily Official Lists*; Grace's Guide = *Grace's Guide to British Industrial History*.

Sources: Balance sheets, SEYB, and SEDOL are available as part of the Stock Exchange Collection at the London Archives, Guildhall Library, London. Grace's Guide is available online (https://www.gracesguide.co.uk/).

APPENDIX C

Additional results and robustness exercises

Table C1. Test of pre-reform trend differences

Variable	(1) Total Debt/Total Assets	(2) Total Debt/Total Assets
Patenting Firm	0.031	0.065
	(0.047)	(0.056)
1900.AccountYear	0.007	0.006
	(0.021)	(0.020)
1901.AccountYear	0.009	0.006
	(0.021)	(0.021)
1902.AccountYear	0.005	0.004
	(0.021)	(0.021)
1903.AccountYear	0.000	0.002
	(0.020)	(0.020)
1904.AccountYear	0.006	0.009
	(0.021)	(0.020)
Patenting#1900	-0.049	-0.062
	(0.059)	(0.071)
Patenting#1901	-0.041	-0.046
	(0.056)	(0.067)
Patenting#1902	-0.029	-0.049
	(0.056)	(0.065)
Patenting#1903	-0.028	-0.062
	(0.053)	(0.063)
Patenting#1904	-0.023	-0.064
	(0.053)	(0.062)
Constant	0.234***	0.233***
	(0.016)	(0.015)
Observations	917	917
R-squared	0.001	0.002

Notes: This table tests for differences in pre-trends for patenting and non-patenting firms, before the introduction of patent examination (1899–1904). The dependent variable is Total Debt/Total Assets. Independent variables include year dummy variables, a dummy variable identifying patenting firms, and year-patenting firm interactions. Column (1) uses the year of patent filing to identify a patenting firm; column (2) uses the year of patent grant to identify a patenting firm. See Table B4 for variable definitions and sources. Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table C2. Pooled analysis

Variable	(1) Total Debt	(2) Total Debt	(3) Bond Debt	(4) Bond Debt	(5) All Non-Bond	(6) All Non-Bond	(7) Bank Debt	(8) Bank Debt	(9) Trade Credit	(10) Trade Credit
Patenting Firm	-0.008	-0.009	-0.017*	-0.017*	0.009	0.008	0.003*	0.003*	0.004	0.004
	(0.009)	(0.009)	(0.009)	(0.009)	(0.006)	(0.006)	(0.002)	(0.002)	(0.005)	(0.005)
Examination Years	0.029	0.015	0.001	-0.002	0.027	0.017	0.027	0.028	-0.006	-0.018
	(0.066)	(0.067)	(0.049)	(0.051)	(0.048)	(0.048)	(0.017)	(0.018)	(0.042)	(0.043)
Patenting x Examination	0.024**	0.024**	0.031**	0.030**	-0.007	-0.006	-0.007**	-0.007**	0.002	0.003
	(0.012)	(0.012)	(0.012)	(0.013)	(0.007)	(0.007)	(0.003)	(0.003)	(0.006)	(0.006)
ROA	-0.393***	-0.433***	-0.274***	-0.287***	-0.119**	-0.146**	-0.069***	-0.067***	-0.037	-0.062
	(0.074)	(0.074)	(0.060)	(0.064)	(0.055)	(0.057)	(0.022)	(0.025)	(0.048)	(0.050)
Total Reserves/Total Assets	-0.466***	-0.459***	-0.285***	-0.270***	-0.181***	-0.189***	-0.052*	-0.053*	-0.135***	-0.143***
	(0.075)	(0.076)	(0.063)	(0.063)	(0.055)	(0.057)	(0.027)	(0.028)	(0.040)	(0.041)
Age	-0.000	0.001	0.000	0.001	-0.000	0.001	-0.002	-0.003	0.003	0.004
	(0.008)	(0.008)	(0.006)	(0.006)	(0.006)	(0.006)	(0.002)	(0.002)	(0.005)	(0.005)
ln (Issued Share Capital)	-0.084***	-0.091***	-0.065***	-0.070***	-0.019	-0.021	-0.009	-0.008	-0.010	-0.012
	(0.022)	(0.022)	(0.023)	(0.024)	(0.013)	(0.013)	(0.007)	(0.007)	(0.009)	(0.010)
Maturity Ratio		0.036		0.020		0.016		-0.006		0.017
		(0.041)		(0.041)		(0.025)		(0.013)		(0.022)
London HQ	-0.179	0.282*	-0.068	0.206*	-0.110	0.077	-0.050*	0.042	-0.051	0.047
	(0.114)	(0.149)	(0.085)	(0.113)	(0.082)	(0.106)	(0.029)	(0.038)	(0.074)	(0.095)
Family Firm	-0.270	-0.318	-0.218	-0.237	-0.052	-0.081	-0.034	-0.028	-0.035	-0.070
	(0.201)	(0.205)	(0.151)	(0.158)	(0.145)	(0.145)	(0.050)	(0.052)	(0.131)	(0.132)
Iron		0.438***		0.264***		0.174***		0.094***		0.082***
		(0.035)		(0.028)		(0.024)		(0.008)		(0.022)
Constant	1.526***	1.174***	1.101***	0.907***	0.424***	0.267	0.223**	0.120	0.190	0.132
	(0.277)	(0.288)	(0.291)	(0.302)	(0.161)	(0.173)	(0.086)	(0.092)	(0.119)	(0.127)
Observations	2,469	2,391	2,469	2,391	2,469	2,391	2,469	2,391	2,469	2,391
R-squared	0.894	0.897	0.922	0.923	0.769	0.768	0.479	0.479	0.785	0.784
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Clustered SE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Notes: This table presents results of pooled regressions examining the introduction of patent examination and patenting firms' debt financing. The dependent variable is Total Debt/Total Assets (columns 1 and 2); Bond Debt/Total Assets (columns 3 and 4); Non-Bond Debt/Total Assets (columns 5 and 6); Bank Debt/Total Assets (columns 7 and 8); Trade Credit Debt/Total Assets (columns 9 and 10). 'Patenting Firm' is a dummy variable that =1 in years once a firm has obtained a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. See Table B4 for variable definitions and sources. Clustered robust standard errors (at the firm level) in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table C3. Robustness to industry-specific trends and sector composition

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Total Debt/Total Assets			Bond Debt/Total Assets		
	Industry trends	Industry x Year FE	Excl. Electricity/ Power	Industry trends	Industry x Year FE	Excl. Electricity/ Power
Patenting Firm	-0.004	-0.004	-0.008	-0.014*	-0.015*	-0.010
	(0.008)	(0.009)	(0.009)	(0.008)	(0.009)	(0.009)
Examination Years	0.044	-0.070***	0.031**	-0.026	-0.060*	0.012
	(0.027)	(0.025)	(0.012)	(0.022)	(0.031)	(0.011)
Patenting x Examination	0.016	0.013	0.026**	0.026***	0.027***	0.026**
	(0.010)	(0.010)	(0.013)	(0.009)	(0.010)	(0.013)
ROA	-0.391***	-0.361***	-0.390***	-0.265***	-0.246***	-0.269***
	(0.067)	(0.065)	(0.071)	(0.053)	(0.052)	(0.057)
Total Reserves/Total Assets	-0.433***	-0.420***	-0.466***	-0.259***	-0.240***	-0.275***
	(0.063)	(0.060)	(0.073)	(0.052)	(0.050)	(0.061)
ln (Issued Share Capital)	-0.082***	-0.073***	-0.082***	-0.060***	-0.055***	-0.061***
	(0.021)	(0.021)	(0.022)	(0.021)	(0.021)	(0.023)
Constant	1.347***	1.337***	1.340***	0.959***	0.968***	0.970***
	(0.268)	(0.268)	(0.283)	(0.273)	(0.268)	(0.299)
Observations	2,469	2,469	2,270	2,469	2,469	2,270
R-squared	0.202	0.309	0.150	0.198	0.291	0.102
Number of Firms	293	293	267	293	293	267
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Clustered SE	YES	YES	YES	YES	YES	YES
Industry-specific trends	YES	NO	NO	YES	NO	NO
Industry x Year FE	NO	YES	NO	NO	YES	NO
Electricity and Power excluded	NO	NO	YES	NO	NO	YES

Notes: This table presents robustness tests addressing industry-specific financing trends and sector composition. Industries are assigned using the 18-category classification developed by Billington and Hanna (2021). The dependent variable is Total Debt/Total Assets in columns (1)–(3) and Bond Debt/Total Assets in columns (4)–(6). Columns (1) and (4) include industry-specific linear trends; columns (2) and (5) include industry-by-year fixed effects; and columns (3) and (6) exclude firms in Electricity and Power. 'Patenting Firm' is a dummy variable that equals one once a firm has obtained a granted *unexamined* patent before 1905 and/or a granted *examined* patent from 1905 onwards. All specifications include firm and year fixed effects and controls for firm profitability, self-financing ability, size, and age. See Table B4 for variable definitions and sources. Clustered robust standard errors at the firm level are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C4. Total debt/total assets analysis after adjusting control sample

Variable	(1) PSM	(2) EB	(3) Never Patent	(4) Continuing Firms
Patenting Firm	-0.011	-0.008		-0.011
	(0.010)	(0.009)		(0.009)
Examination Years	-0.121	-0.029	0.061	0.064
	(0.161)	(0.161)	(0.108)	(0.116)
Patenting x Examination	0.022*	0.025**	0.026**	0.026**
	(0.012)	(0.012)	(0.012)	(0.012)
ROA	-0.382***	-0.424***	-0.378***	-0.386***
	(0.115)	(0.113)	(0.073)	(0.075)
Total Reserves/Total Assets	-0.480***	-0.522***	-0.478***	-0.494***
	(0.106)	(0.117)	(0.077)	(0.075)
Age	0.012	0.006	-0.002	-0.003
	(0.012)	(0.012)	(0.008)	(0.009)
ln (Issued Share Capital)	-0.083***	-0.097***	-0.093***	-0.086***
	(0.028)	(0.027)	(0.023)	(0.022)
Constant	1.282***	1.528***	1.510***	1.440***
	(0.382)	(0.357)	(0.302)	(0.296)
Observations	1,203	2,213	2,105	2,127
Number of Firm	118	227	215	198
R-squared	0.155	0.167	0.152	0.150
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Clustered SE	YES	YES	YES	YES

Notes: This table presents robustness tests examining the sensitivity of patenting firms' debt financing to alternative comparison samples. The dependent variable is Total Debt/Total Assets. Column (1) uses firm-level propensity-score matching (PSM) based on pre-reform profitability, reserves, and size. Matching is conducted one-to-one without replacement, restricted to common support and firms observed before and after 1905. Column (2) uses entropy balancing (EB) to reweight controls based on pre-treatment characteristics. Column (3) restricts the control group to firms that never patented. Column (4) restricts the sample to firms observed both before and after the introduction of patent examination in 1905. 'Patenting Firm' is a dummy variable that =1 in years once a firm has a granted *unexamined* patent before 1905, and a granted *examined* patent from 1905 onwards. 'Examination Years' =1 from 1905 onwards, identifying the new patent examination regime. Control variables for firm profitability, self-financing ability, size and age are included, as are firm and year controls. See Table B4 for variable definitions and sources. Clustered standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table C5. Analysis using alternative firm profitability definition (ROE)

Variable	(1) Total Debt/ Total Assets	(2) Bond Debt/ Total Assets	(3) Other Debt/ Total Assets	(4) Bank Debt/ Total Assets	(5) Trade Credit/ Total Assets
Patenting Firm	-0.009	-0.018**	0.009	0.003*	0.004
	(0.008)	(0.009)	(0.005)	(0.002)	(0.005)
Examination Years	0.031	0.006	0.025	0.038	-0.022
	(0.099)	(0.074)	(0.072)	(0.025)	(0.064)
Patenting x Examination	0.025**	0.032***	-0.007	-0.007**	0.002
	(0.011)	(0.012)	(0.006)	(0.003)	(0.005)
ROE	-0.151***	-0.097***	-0.054*	-0.034***	-0.015
	(0.032)	(0.023)	(0.027)	(0.012)	(0.024)
Total Reserves/Total Assets	-0.458***	-0.278***	-0.180***	-0.051**	-0.135***
	(0.070)	(0.059)	(0.051)	(0.026)	(0.037)
Age	0.000	0.001	-0.000	-0.002	0.003
	(0.008)	(0.006)	(0.005)	(0.002)	(0.005)
ln (Issued Share Capital)	-0.087***	-0.066***	-0.021*	-0.009	-0.010
	(0.021)	(0.022)	(0.012)	(0.007)	(0.009)
Constant	1.399***	1.024***	0.375**	0.160*	0.196
	(0.275)	(0.286)	(0.164)	(0.085)	(0.121)
Observations	2,469	2,469	2,469	2,469	2,469
R-squared	0.134	0.099	0.049	0.027	0.031
Number of Firms	293	293	293	293	293
Year FE	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES
Clustered SE	YES	YES	YES	YES	YES

Notes: This table replicates key results from the paper's main analysis, but substituting Return on Equity ('ROE') for Return on Assets ('ROA'). 'Patenting Firm' is a dummy variable that =1 in years once a firm has obtained a granted *unexamined* patent before 1905, and/or a granted *examined* patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. See Table B4 for variable definitions and sources. Clustered robust standard errors (at the firm level) in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table C6. Analysis using patent filing date rather than grant date

Variable	(1) Total Debt/ Total Assets	(2) Bond Debt/ Total Assets	(3) Other Debt/ Total Assets	(4) Bank Debt/ Total Assets	(5) Trade Credit/ Total Assets
Patenting	-0.011	-0.022**	0.012**	0.004*	0.007
	(0.009)	(0.009)	(0.006)	(0.002)	(0.005)
Examination Years	0.038	0.011	0.027	0.038	-0.020
	(0.099)	(0.074)	(0.072)	(0.025)	(0.064)
Patenting x Examination	0.026**	0.033***	-0.007	-0.004	-0.001
	(0.011)	(0.012)	(0.006)	(0.003)	(0.006)
ROA	-0.391***	-0.270***	-0.121**	-0.069***	-0.039
	(0.070)	(0.056)	(0.051)	(0.021)	(0.045)
Total Reserves/Total Assets	-0.466***	-0.284***	-0.183***	-0.052**	-0.136***
	(0.070)	(0.059)	(0.051)	(0.025)	(0.037)
Age	-0.000	-0.000	-0.000	-0.002	0.003
	(0.007)	(0.006)	(0.005)	(0.002)	(0.005)
ln (Issued Share Capital)	-0.084***	-0.065***	-0.020	-0.009	-0.010
	(0.021)	(0.022)	(0.012)	(0.007)	(0.009)
Constant	1.385***	1.022***	0.363**	0.154*	0.192
	(0.275)	(0.280)	(0.163)	(0.084)	(0.121)
Observations	2,469	2,469	2,469	2,469	2,469
R-squared	0.144	0.112	0.050	0.025	0.031
Number of Firms	293	293	293	293	293
Year FE	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES
Clustered SE	YES	YES	YES	YES	YES

Notes: This table replicates key results from the paper's main analysis, but using patent filing date rather than grant date in judging whether a firm was affected by the new patent examination regime. 'Patenting Firm' is a dummy variable that =1 in years once a firm has filed a patent before 1905, and/or a filed patent from 1905 onwards. 'Examination Years' is a dummy variable that =1 from 1905 onwards, identifying the new patent examination regime. See Table B4 for variable definitions and sources. Clustered robust standard errors (at the firm level) in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table C7. Staggered difference-in-differences estimates: Total debt / total assets

Panel A. Aggregate treatment effects

Specification	ATT	SE	p-value	95% CI
Primary C&S	0.0232	0.0109	0.034	[0.0018, 0.0446]
Balanced event-time sample, $e = 0$ to $+5$	0.0218	0.0098	0.026	[0.0026, 0.0410]
Never-treated controls only	0.0232	0.0109	0.033	[0.0019, 0.0446]

Panel B. Event-time treatment effects

Event time	Primary				Balanced			
	ATT	Pointwise p-value	95% Pointwise CI	95% Simultaneous CI	ATT	Pointwise p-value	95% Pointwise CI	95% Simultaneous CI
-5	0.0086	0.446	[−0.0135, 0.0307]	[−0.0200, 0.0372]	−0.0036	0.832	[−0.0368, 0.0296]	[−0.0466, 0.0395]
-4	−0.0145	0.197	[−0.0366, 0.0076]	[−0.0431, 0.0140]	−0.0232	0.141	[−0.0539, 0.0076]	[−0.0631, 0.0168]
-3	−0.0003	0.976	[−0.0216, 0.0209]	[−0.0277, 0.0271]	−0.0043	0.807	[−0.0386, 0.0300]	[−0.0487, 0.0401]
-2	0.0025	0.722	[−0.0113, 0.0164]	[−0.0154, 0.0204]	0.0071	0.511	[−0.0141, 0.0283]	[−0.0204, 0.0346]
-1			*Reference period*				*Reference period*	
0	0.012	0.072	[−0.0011, 0.0251]	[−0.0049, 0.0289]	0.0071	0.404	[−0.0096, 0.0238]	[−0.0145, 0.0288]
1	0.0098	0.279	[−0.0079, 0.0274]	[−0.0130, 0.0326]	0.0033	0.777	[−0.0196, 0.0262]	[−0.0264, 0.0330]
2	0.0121	0.171	[−0.0052, 0.0294]	[−0.0102, 0.0344]	0.0182	0.138	[−0.0059, 0.0422]	[−0.0130, 0.0493]
3	0.0088	0.545	[−0.0198, 0.0374]	[−0.0281, 0.0457]	0.0211	0.274	[−0.0167, 0.0588]	[−0.0278, 0.0699]
4	0.0257	0.045	[0.0006, 0.0507]	[−0.0066, 0.0580]	0.0282	0.049	[0.0002, 0.0562]	[−0.0081, 0.0644]
5	0.0531	0.01	[0.0128, 0.0935]	[0.0011, 0.1052]	0.0531	0.01	[0.0128, 0.0935]	[0.0009, 0.1054]

Panel C. Dynamic summary

Summary measure	Primary			Balanced		
	ATT	p-value	95% CI	ATT	p-value	95% CI
Average ATT, 0 to $+3$	0.0107	0.037	[0.0006, 0.0207]	0.0124	0.065	[−0.0008, 0.0256]
Average ATT, $+4$ to $+5$	0.0394	0.001	[0.0157, 0.0631]	0.0407	0.001	[0.0161, 0.0652]
Difference: $+4/+5 - 0/+3$	0.0287	0.029	[0.0029, 0.0545]	0.0282	0.047	[0.0004, 0.0561]

Notes: The table reports Callaway and Sant'Anna (2021) staggered difference-in-differences estimates for the ratio of total debt to total assets. Treatment begins when a firm first receives an examined patent from 1905 onwards. The primary specification uses not-yet-treated and never-treated firms as comparisons, with pairwise balancing for the unbalanced panel and 9,999 multiplier-bootstrap repetitions.

Panel A reports aggregate treatment effects on the treated (ATTs). The primary estimate is the simple overall ATT; the balanced estimate averages event times 0 to +5 for cohorts observed throughout that window; and the never-treated specification excludes not-yet-treated comparisons.

Panel B reports event-time ATTs relative to event time −1, the year before treatment, which is normalised to zero. Effects are not cumulative. Balanced estimates use cohorts observed through five post-treatment years, limiting changes in cohort composition across event time. Pointwise p-values and confidence intervals evaluate each event time separately, whereas simultaneous confidence intervals cover the full event-study profile and are therefore more conservative.

Panel C reports average ATTs over event times 0 to +3 and +4 to +5, followed by the difference between those averages. Standard errors for these summaries are calculated using the full covariance matrix of the event-time estimates.

Table C8. Summary statistics from placebo simulations

Statistic	Value
Number of simulations	1,000
Mean placebo 'examined patent' coefficient ($\hat{\beta}$)	−0.003
Standard deviation of placebo coefficient	0.013
% of significant placebo effects ($p < 0.05$)	6.60%
% of significant and positive placebo effects	1.3% (13/1000)

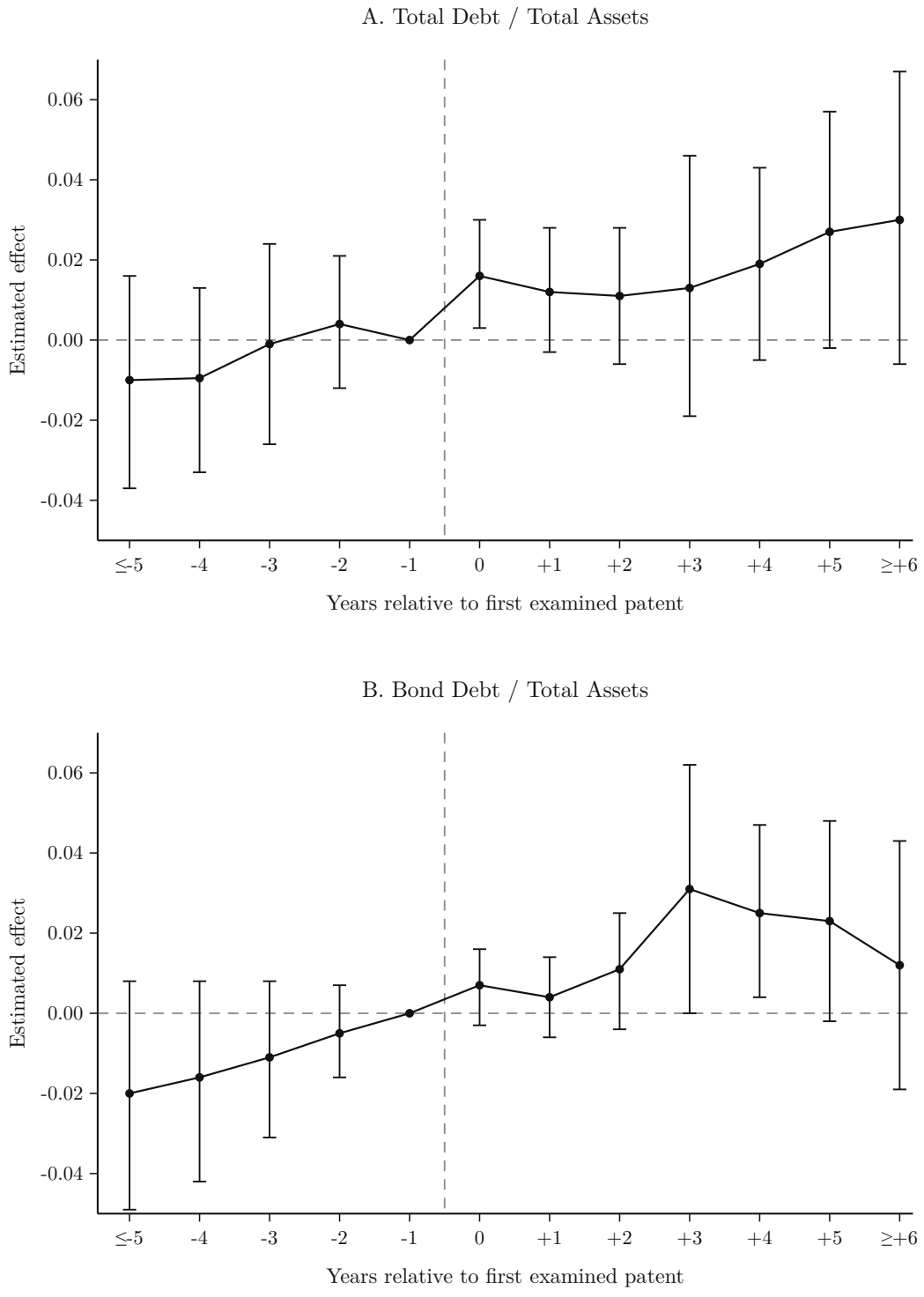
Notes: This table summarizes the results from 1,000 placebo simulations designed to assess the robustness of the estimated treatment effect of patent examination on firm leverage. In each iteration, patenting status and patent timing were randomly assigned at the firm level. All placebo regressions replicate the specification of the main difference-in-differences model in Table 5, column 2.

Table C9. Contrast analysis, pre- versus post-reform

Aspect	Pre-Reform (Before 1905)	Post-Reform (After 1905)	Key Insight
Patent Application Process	Patents granted following a simple formalities examination, with a low fee.	Applications subjected to prior-art examination, with £1 fee increase.	Examination created a non-price barrier that made it harder for weak claims to proceed unchanged to sealing.
Role of Patent Examiners	No public investigation of prior art beyond limited formalities checks.	Technically trained examiners searched earlier British patents and reported potentially anticipatory claims.	Examination introduced standardised state verification and public prior-art information without full substantive examination.
Patent Application Abandonment Rates	Applications were rarely abandoned because patent claims faced little scrutiny before sealing.	Withdrawal and amendment increased after applicants received adverse prior-art information.	Selection occurred after the search as applicants responded to the information it produced.
Patent Quality	Strong and weak claims remained pooled, limiting the information conveyed by registration.	Withdrawal and amendment increased the average strength of patents proceeding to sealing.	Examination improved average patent quality without guaranteeing validity, enforceability, or commercial success.
Signal to Investors	Registration provided no common public basis for distinguishing strong from weak claims.	The public examination record informed investors about technical and economic credibility and the probability of effective legal protection.	Technical and economic credibility and effective legal protection jointly determined expected pledgeable value. Their effects cannot be separated empirically.
Debt Accessibility for Firms	Patenting firms showed similar debt-to-assets ratios to non-patenting firms.	Patenting firms achieved higher debt-to-assets ratios, driven by bonds, with no comparable rise in bank borrowing or fall in coupons.	The public state signal was especially valuable to dispersed bond investors, expanding bond-market capacity at prevailing terms.
Investment and Firm Growth	Financing constraints could prevent firms from commercialising inventions or undertaking subsequent R&D.	Greater bond capacity enabled previously constrained firms to finance additional investment and grow.	Examination could support commercialisation, subsequent R&D, and firm growth where financing constraints had previously bound.

Notes: This table summarises the institutional differences between the pre- and post-reform patent systems, the mechanisms developed in the model, and the principal empirical findings.

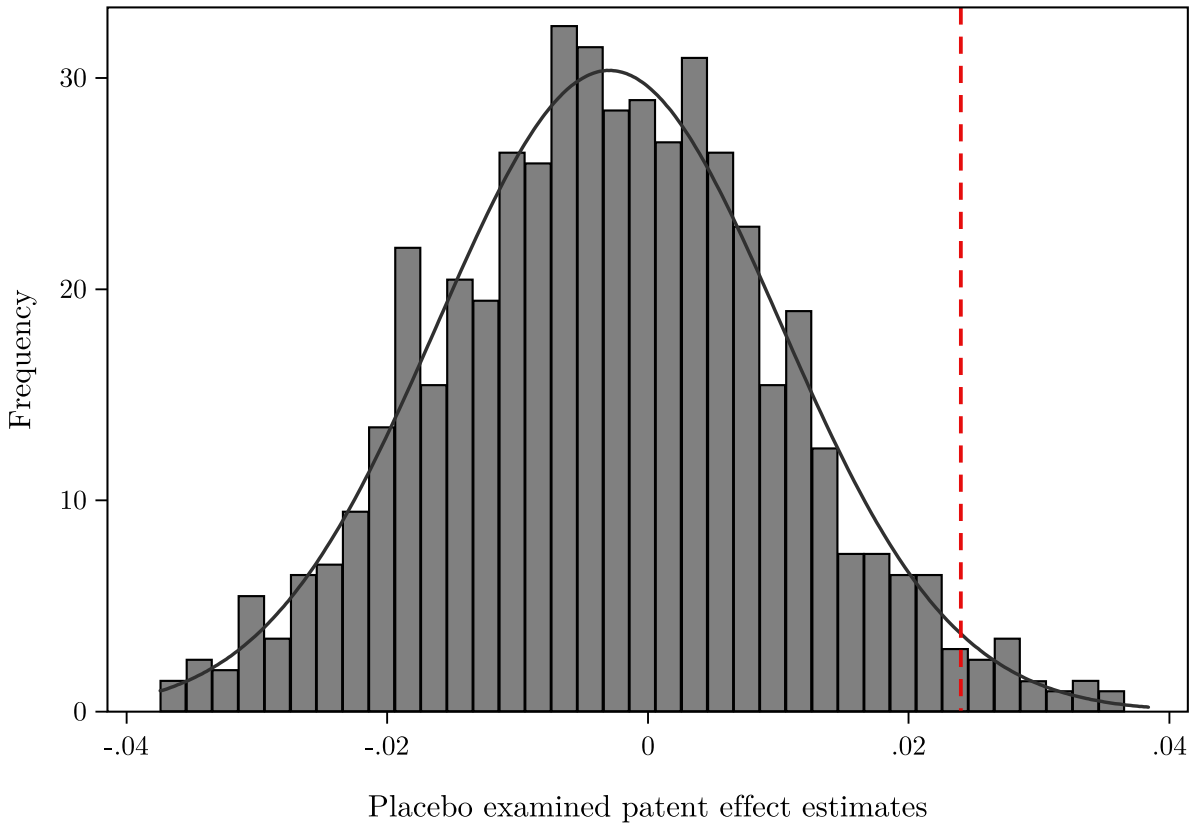
Figure C1. Dynamic effects of patent examination on firm borrowing



Notes: This figure reports event-study estimates of the effect of receiving a first examined patent on firms' borrowing. The dependent variable is Total Debt/Total Assets in Panel A and Bond Debt/Total Assets in Panel B. Event time is measured relative to the year in which the firm received its first examined patent. The year immediately preceding treatment, $t = -1$, is the omitted reference period. Observations five or more years before treatment and six or more years after treatment are grouped into the endpoint categories. All specifications include firm and year fixed effects and controls for profitability (ROA), self-financing capacity (Total Reserves/Total Assets), age, and size. Vertical bars report 95% confidence intervals based on standard errors clustered at the firm level. The vertical dashed line marks the first examined patent; the horizontal dashed line marks zero.

Sources: Authors' calculations using data described in Section 4.

Figure C2. Distribution of placebo estimates from simulated regressions



Notes: This figure presents the distribution of estimated examined-patent effects from 1,000 placebo difference-in-differences regressions. In each simulation, firms are randomly assigned a placebo patenting-firm indicator and patenting year. Each regression uses the firm's total debt-to-assets ratio and replicates the specification in Table 5, column (2), including firm and year fixed effects, controls for profitability (ROA), liquidity (Total Reserves/Total Assets), age, and size, and standard errors clustered at the firm level. The vertical red dashed line marks the actual estimate from this specification (0.024).